Sri Lanka Institute of Information Technology

Final Document

Web-Based Career Selection and Education Guidance System

for Sri Lankan School Leavers After O/L and A/L Examinations

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DECLARATION

We, the undersigned, hereby declare that the work presented in this report is our own original work and has not been submitted previously for any degree or examination. All sources used have been duly acknowledged.

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ABSTRACT

Career selection and educational planning are among the most significant challenges faced by Sri Lankan school leavers after completing their G.C.E. Ordinary Level (O/L) and Advanced Level (A/L) examinations. The lack of structured, accessible, and personalized guidance often leads to uninformed decisions, unnecessary stress, and missed opportunities. This project presents the design and development of a Web-Based Career Selection and Education Guidance System specifically tailored to address these challenges for Sri Lankan students.

The system is built on a full-stack MERN architecture comprising MongoDB, Express.js, React.js, and Node.js. It integrates five core functional modules: a Career Assessment Module that evaluates student interests through structured quizzes; a Qualification and Eligibility Analyzer that uses historical Z-score cut-off data to determine university program eligibility; a Career Path Explorer that provides detailed information on a wide range of career options; an Alternative Pathway Recommender that suggests private degrees, diplomas, and vocational training routes; and an Admin and Job Management System for content administration and job listing management.

The system supports role-based access control for students and administrators, Google OAuth integration, and a responsive user interface that works across devices. Evaluation through functional testing and user feedback demonstrates that the system achieves its core objectives of improving career awareness, reducing decision-related uncertainty, and expanding access to structured guidance for Sri Lankan school leavers.

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We would also like to thank the Sri Lankan students who participated in user testing and provided honest feedback that helped shape the final design of this system.

Finally, we are grateful to our families for their patience and unwavering moral support throughout this academic journey.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
A/L	Advanced Level
API	Application Programming Interface
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
EER	Enhanced Entity-Relationship
HTML	Hypertext Markup Language
HTTP/HTTPS	Hypertext Transfer Protocol / Secure
IT	Information Technology
JWT	JSON Web Token
MERN	MongoDB, Express.js, React.js, Node.js
MVC	Model-View-Controller
O/L	Ordinary Level
OAuth	Open Authorization
REST	Representational State Transfer
SLIIT	Sri Lanka Institute of Information Technology
SWOT	Strengths, Weaknesses, Opportunities, Threats
UGC	University Grants Commission
UI	User Interface
UX	User Experience
UML	Unified Modelling Language

Chapter 1: Introduction

1.1 Problem and Motivation

Career selection and educational planning represent two of the most consequential decisions that Sri Lankan school leavers must make after completing their G.C.E. Ordinary Level (O/L) and Advanced Level (A/L) examinations. These decisions significantly shape their future academic trajectories, professional prospects, and overall quality of life. Despite the gravity of these choices, a large proportion of students navigate this process without adequate structured support.

The current guidance landscape in Sri Lanka is predominantly examination-result oriented. School career guidance programmes, where they exist, tend to focus narrowly on government university admission requirements, leaving many students unaware of the broader range of educational and vocational pathways available to them. Students who do not meet government university Z-score cut-offs often feel that their options are exhausted, when in reality, numerous viable alternatives exist, including private degree programmes, professional certifications, diplomas, and vocational training qualifications.

Furthermore, the growing complexity of the modern labour market means that career awareness must extend beyond academic credentials. Students require information about required skills, industry demand levels, salary ranges, and career progression paths. Without a centralized, accessible, and personalized platform, this information remains fragmented and difficult for school leavers to access and act upon.

This project is motivated by the need to bridge this guidance gap by developing a web-based system that consolidates career information, eligibility analysis, and pathway recommendations into a single, student-friendly interface.

1.2 Literature Review

Several international and local studies have explored the role of technology in career guidance. Research by Srivastava and Bing (2019) demonstrated that web-based career guidance systems significantly improve decision-making confidence among secondary school students by providing structured, data-driven recommendations. In the Sri Lankan context, studies have highlighted that the majority of school leavers rely on informal channels such as family advice and peer discussion, which are often limited in scope and accuracy.

Existing international platforms such as Careers360 (India), MyFuture (Australia), and the Morrisby Platform (UK) demonstrate that combining interest-based assessments with qualification analysis can meaningfully improve the quality of career decision-making. However, these platforms are not localized for the Sri Lankan educational system, which has unique characteristics including the Z-score-based university admission system and the distinction between O/L and A/L academic streams.

Prior work on recommender systems in education, including collaborative filtering and content-based filtering approaches (Drachsler et al., 2015), provides a theoretical basis for personalized pathway recommendations. The integration of such techniques with local eligibility data offers a promising approach for the Sri Lankan context. This project draws on these foundations to develop a system that is both technically robust and contextually relevant.

1.3 Aim and Objectives

The aim of this project is to design, develop, and evaluate a web-based career selection and education guidance system that supports Sri Lankan school leavers in making informed decisions about their post-examination educational and career pathways.

To achieve this aim, the following specific objectives are defined:

- To develop a platform that enables students to explore career options with detailed information on required qualifications, skills, and future prospects.
- To implement a qualification and eligibility analyzer that uses historical Z-score cut-off data to determine likely eligibility for government university programmes.
- To provide personalized career recommendations based on interest and skill assessments.
- To suggest alternative education pathways including private degrees, diplomas, and vocational training programmes when students do not qualify for their preferred government university courses.
- To support side-by-side comparison of career and education options to aid informed decision-making.
- To develop an administrative interface for content managers to manage career data, eligibility information, and job or internship listings.

1.4 Solution Overview

The proposed system is a full-stack web application built using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The system is structured into five core functional modules: the Career Assessment Module, the Qualification and Eligibility Analyzer, the Career Path Explorer, the Alternative Pathway Recommender, and the Admin and Job Management System. These modules are accessed through a responsive React-based front end and are supported by a RESTful Node.js/Express.js back end and a MongoDB database.

The system supports two primary user roles: students and administrators. Students can register, complete interest-based assessments, check eligibility for university programmes, explore careers, compare pathways, and view job and internship listings. Administrators can manage all content through a secure admin dashboard.

The Git repository for this project is available at:

https://github.com/IT24102099/A-Web-Based-Career-Selection-and-Education-Guidance-System

Chapter 2: Requirement Analysis

2.1 Stakeholder Analysis

A stakeholder analysis was conducted to identify all parties who have an interest in, or are affected by, the system. Understanding stakeholder needs and expectations is essential to ensuring that the system delivers value and meets real-world requirements.

2.1.1 Primary Stakeholders

- School Leavers / Students (O/L and A/L): The primary end users of the system. They register on the platform, enter their examination results, complete career assessments, and use the system to explore career and education options. Their needs include personalized guidance, clear eligibility information, and accessible pathway comparison.
- Content Managers / Career Guidance Administrators: Responsible for maintaining the accuracy and currency of career information, Z-score cut-off data, course listings, and quiz content. They require an intuitive admin interface with full CRUD capabilities.
- System Administrators / IT Administrators: Responsible for managing user accounts, roles, system security, data backups, and overall platform maintenance.

2.1.2 Secondary Stakeholders

- Academic and Career Counsellors: May use the system to support their guidance work, reviewing student profiles and recommendations.
- Educational Institutions and Private Providers: Benefit from the system by having their programmes listed and recommended to eligible students.
- Parents and Guardians: Indirectly benefit from improved student decision-making and access to structured guidance.

2.2 Feasibility and SWOT Analysis

2.2.1 Feasibility Analysis

Technical Feasibility: The system is developed using the MERN stack, which is a widely used and well-documented technology combination. All required tools and libraries are open-source and freely available. The team has the necessary skills and experience to deliver the system within the project timeline.

Operational Feasibility: The system addresses a clearly identified need among Sri Lankan school leavers. The web-based delivery model ensures accessibility from any internet-connected device. The admin interface is designed to be usable by non-technical content managers.

Economic Feasibility: The system is developed using open-source technologies and can be hosted on low-cost cloud platforms. In the initial phase, free-tier hosting is sufficient. The system requires no hardware procurement beyond development machines.

Schedule Feasibility: The planned development timeline of twelve weeks is sufficient to design, implement, test, and document the system given the team size and scope.

2.2.2 SWOT Analysis

Strengths:

- Centralized, locally relevant career guidance platform tailored to the Sri Lankan educational context.
- Combines multiple guidance functions (assessment, eligibility, exploration, alternative pathways) in a single system.
- Role-based access control ensures appropriate data access and security.
- MERN stack provides a scalable and maintainable architecture.

Weaknesses:

- Eligibility analysis accuracy depends on the completeness and currency of stored Z-score data.
- Career recommendations rely on quiz responses which may not fully capture student aptitude.
- Initial data population requires significant manual effort.

Opportunities:

- Significant unmet demand for localized, digital career guidance in Sri Lanka.
- Potential for integration with UGC data sources and educational ministry databases.
- Future expansion to mobile application and AI-driven recommendation enhancements.

Threats:

- Rapid changes in university admission policies may make stored eligibility data outdated.
- Low internet connectivity in rural areas may limit access.
- Competition from informal guidance channels and social media.

2.3 Requirements Modelling

2.3.1 Functional Requirements

The following functional requirements were identified through stakeholder consultation and domain analysis:

- FR1: The system shall allow students to register using email/password or Google OAuth and manage their profiles.
- FR2: The system shall provide a structured career interest assessment quiz with configurable questions and scoring logic.
- FR3: The system shall allow students to enter their examination stream, district, Z-score, and year to check eligibility for government university degree programmes.
- FR4: The system shall display eligible degree programmes with probability labels, cut-off comparisons, and trend graphs.
- FR5: The system shall provide a searchable and filterable career explorer with salary ranges, demand levels, and detailed career information pages.
- FR6: The system shall recommend alternative education pathways (private degrees, diplomas, vocational training) based on student profiles and preferences.

- FR7: The system shall support side-by-side comparison of alternative pathways.
- FR8: The system shall provide a job and internship listing page with filtering by type and work mode.
- FR9: Administrators shall be able to manage careers, courses, quiz questions, eligibility data, jobs, universities, and users through a secure admin dashboard.
- FR10: The system shall implement role-based access control distinguishing student and admin roles.

2.3.2 Non-Functional Requirements

- NFR1 – Performance: Key pages shall load within three seconds under standard network conditions.
- NFR2 – Security: All data transmission shall use HTTPS. User passwords shall be hashed using bcrypt. JWT tokens shall be used for session management.
- NFR3 – Usability: The interface shall be intuitive for users with basic digital literacy. Clear error messages and guidance shall be provided throughout.
- NFR4 – Scalability: The system architecture shall support increasing user load, particularly during peak periods following examination result releases.
- NFR5 – Compatibility: The system shall function correctly on current versions of Chrome, Firefox, Safari, and Edge, across desktop and mobile devices.
- NFR6 – Maintainability: The codebase shall follow modular design principles to facilitate future updates and feature additions.

Chapter 3: Design and Development

3.1 System Architecture

The system is designed using a three-tier architecture comprising a presentation layer, an application layer, and a data layer. This separation of concerns ensures modularity, scalability, and maintainability.

The presentation layer is built with React.js and is responsible for rendering the user interface and managing user interactions. The application layer is implemented using Node.js with Express.js and handles all business logic, API request processing, authentication, eligibility calculations, and recommendation generation. The data layer uses MongoDB as the primary database, accessed via the Mongoose ODM.

The three tiers communicate via a RESTful HTTPS API. All client requests are routed through the Express.js server, which validates inputs, processes business logic, and queries or updates the MongoDB database as required. External services for email notifications and file storage are planned as optional future extensions.

Figure 3.1 illustrates the MERN Stack system architecture showing the interactions between the presentation layer, the application layer services, and the MongoDB database.

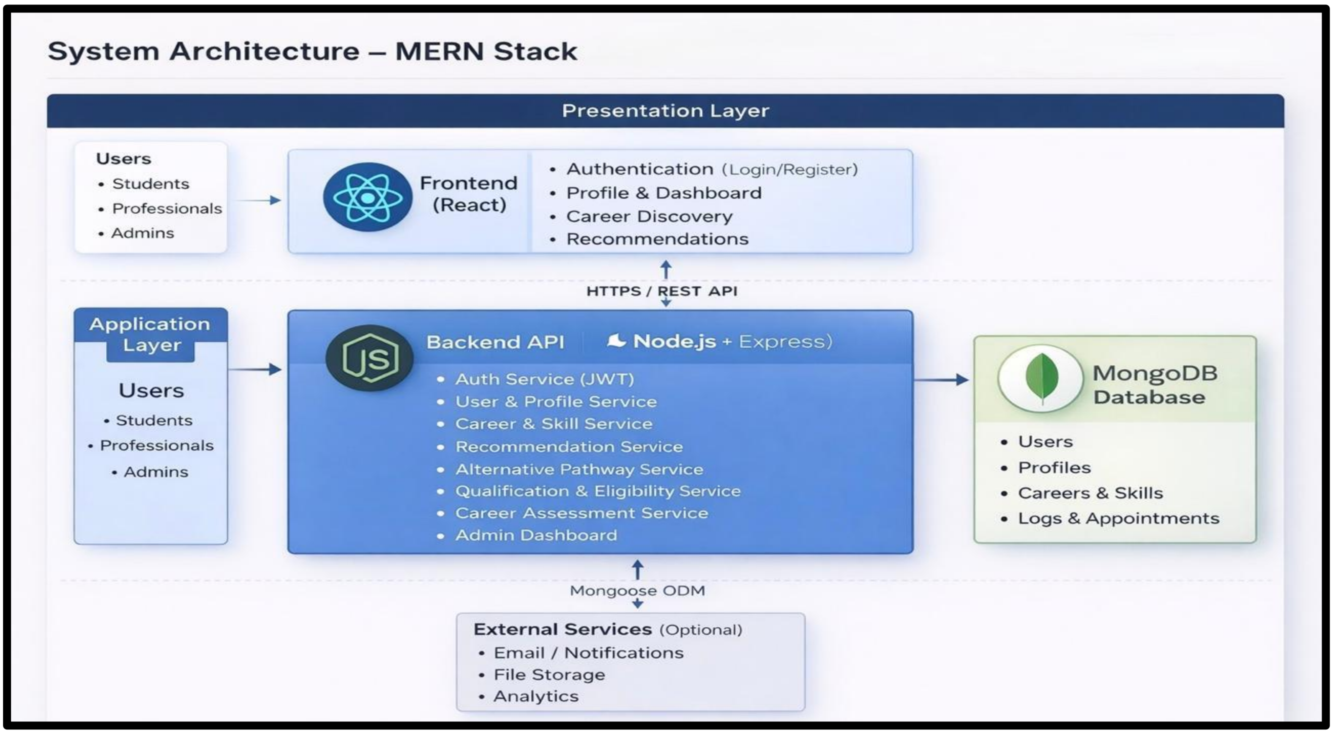

Figure 3.1: MERN Stack System Architecture Diagram

3.2 Component and Module Design

The application is organized into five core functional modules, each responsible for a distinct area of system functionality.

3.2.1 Career Assessment Module and GitHub Activity View

This module is developed by Sithirasenan Vipooshan (IT24102869). It provides an interest and skill-based assessment quiz to help students identify suitable career paths. The module supports stream-based question filtering (Maths, Bio, Commerce, Arts, Tech, or All Streams) and a configurable set of multiple-choice questions. Scoring logic aggregates student responses across career categories and produces a ranked list of top career matches with a score breakdown. Administrators can manage quiz questions through the admin dashboard. Additionally implemented a GitHub-based project activity view to support transparency of team development work. This feature provides an overview of repository activity including commits and contributions, allowing users and evaluators to understand development progress. It enhances project visibility and demonstrates version control practices followed during the development process.

3.2.2 Qualification and Eligibility Analyzer

This module is developed by Naguleswaran Mathuppriya (IT24102099). It allows students to enter their A/L stream, district, Z-score, and examination year to check eligibility for government university degree programmes. The module compares the student's Z-score against stored historical cut-off data and calculates a probability label (High, Medium, or Low) for each eligible programme. A trend graph visualizes Z-score cut-off changes over the past three years. Students can save their eligibility check results and view their history.

3.2.3 Career Path Explorer

This module is developed by Mariyappan Thamilpiriyan (IT23296800). It provides a searchable and filterable career listing page displaying careers with associated images, salary ranges, demand levels, and relevant educational streams. Students can filter by stream, salary range, and demand level. Detailed career pages include descriptions, work environments, education paths, and required skills. Students can save careers to their personal lists and compare multiple careers side by side.

3.2.4 Alternative Pathway Recommender

This module is developed by Kamshiga Ganesan (IT24101365). It presents alternative education and training options including private degrees, diplomas, vocational programmes, foundation courses, and online certifications. A personalized recommendation engine allows students to input their A/L stream, district, budget, preferred duration, government university eligibility status, and career interests to receive a ranked list of up to ten pathway recommendations. A comparison feature allows side-by-side evaluation of selected pathways across key attributes such as provider, type, duration, fees, rating, stream tags, and entry requirements.

3.2.5 Admin and Job Management System

This module is developed by Moosika Ramanathan (IT24102839). It provides a secure admin dashboard with panels for managing users, careers, courses, quiz questions, jobs, universities, and degree programme eligibility data. The dashboard displays summary statistics and recent activity. Job and internship listings include title, company, location, salary, work mode, application deadline, and description. Listings can be filtered by type and mode on the public-facing jobs page.

3.3 Process and Workflow Diagrams

The Use Case Diagram (Figure 3.2) illustrates the full set of interactions between the two primary actors (User/Student and Admin) and the system. Key student use cases include login, registration, Google OAuth login, taking the career quiz, checking eligibility, browsing careers and pathways, and saving careers. Admin use cases include managing careers, courses, quiz questions, jobs, universities, eligibility data, and user accounts.

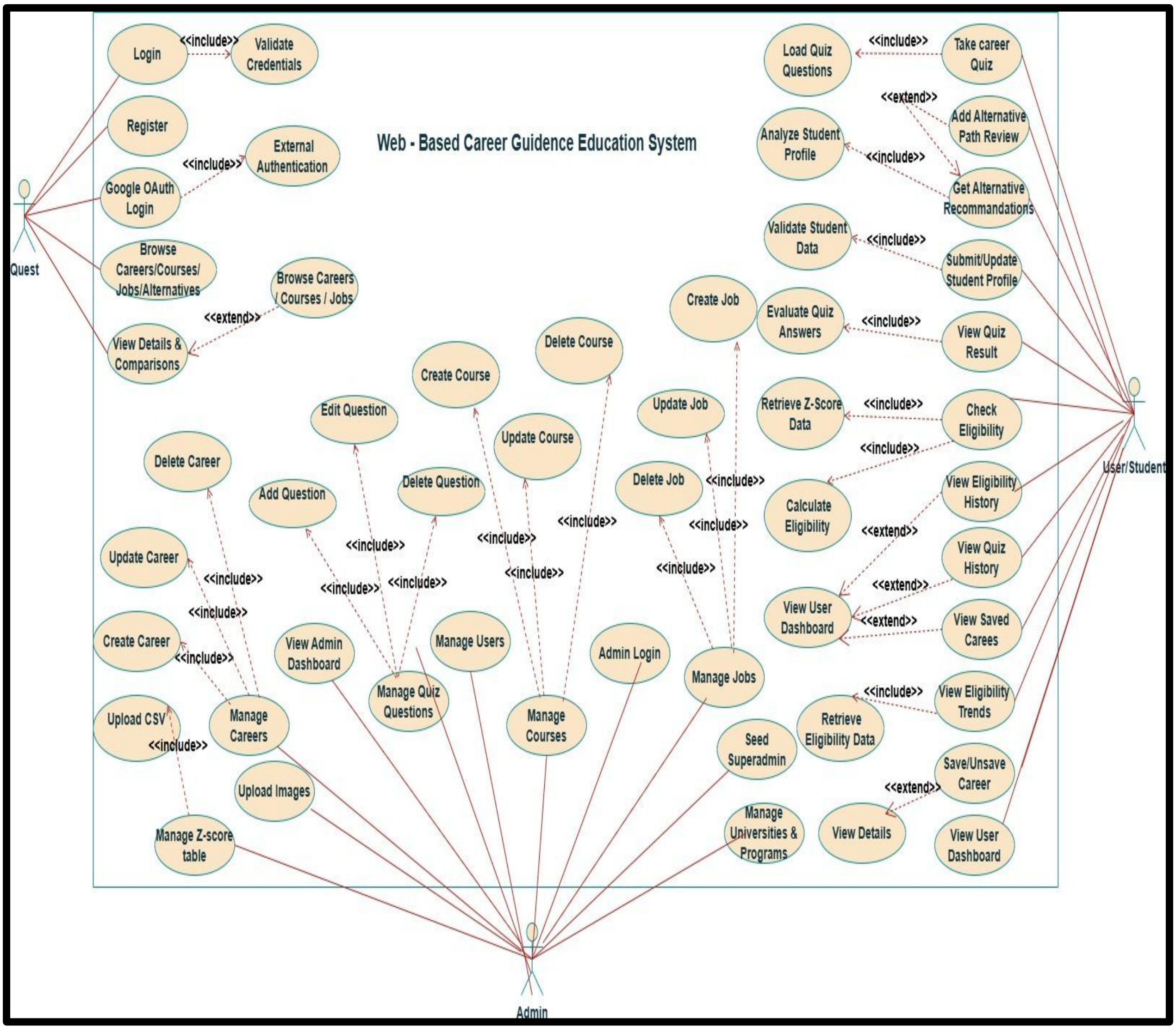

Figure 3.2: Use Case Diagram – Web-Based Career Guidance Education System showing User/Student and Admin actors with all associated use cases and include/extend relationships

The Data Flow Diagram (Figure 3.3) shows the flow of data between students, administrators, the MERN backend, the MongoDB database, and supporting services including the recommendation engine and analytics dashboard. It illustrates how API calls facilitate CRUD operations for all major data entities including users, profiles, careers, assessments, eligibility records, and logs.

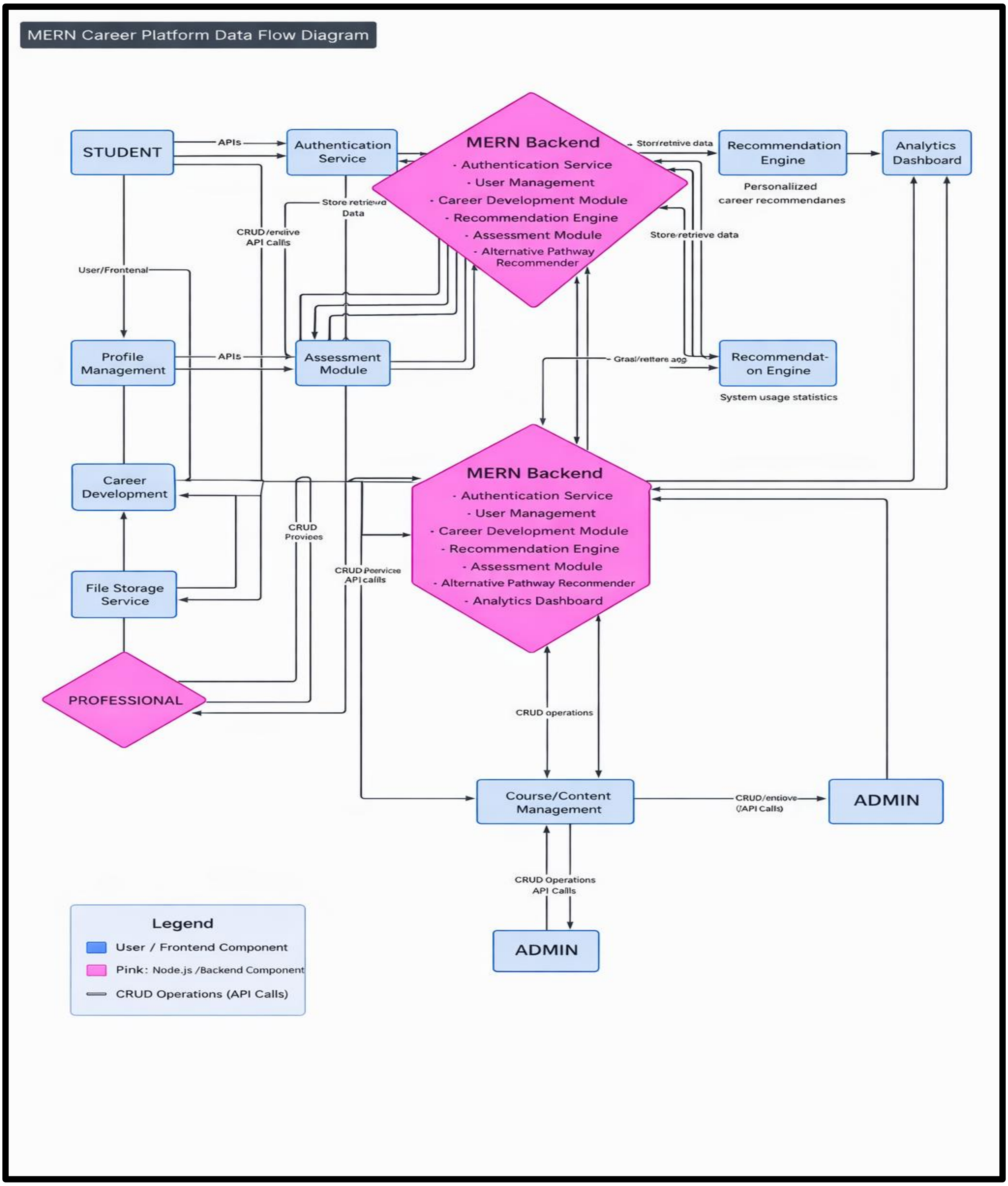

Figure 3.3: MERN Career Platform Data Flow Diagram – showing data flows between Student, Professional, Admin, MERN Backend, MongoDB Database, Recommendation Engine, and Analytics Dashboard

3.4 Database Design

The system uses MongoDB as its primary database, accessed via the Mongoose ODM. The database schema is designed to support all five modules and their interdependencies. Key entities include USER, STUDENT, ADMIN, CAREER, ALTERNATIVEPATH, COURSE, QUIZQUESTION, QUIZANSWER, QUIZRESULT, ELIGIBILITYCHECK, ELIGIBILITYRESULT, ZSCORETABLE, DEGREEPROGRAM, UNIVERSITY, JOB, and SAVEDCAREER.

The Enhanced Entity-Relationship (EER) Diagram (Figure 3.4) illustrates the relationships and attributes of these entities. Notable relationships include the USER entity having a one-to-many relationship with QUIZRESULT and ELIGIBILITYCHECK records, and a many-to-many relationship with SAVEDCAREER. The DEGREEPROGRAM entity is linked to the ZSCORETABLE for cut-off data and to the ELIGIBILITYRESULT entity for computed eligibility outputs.

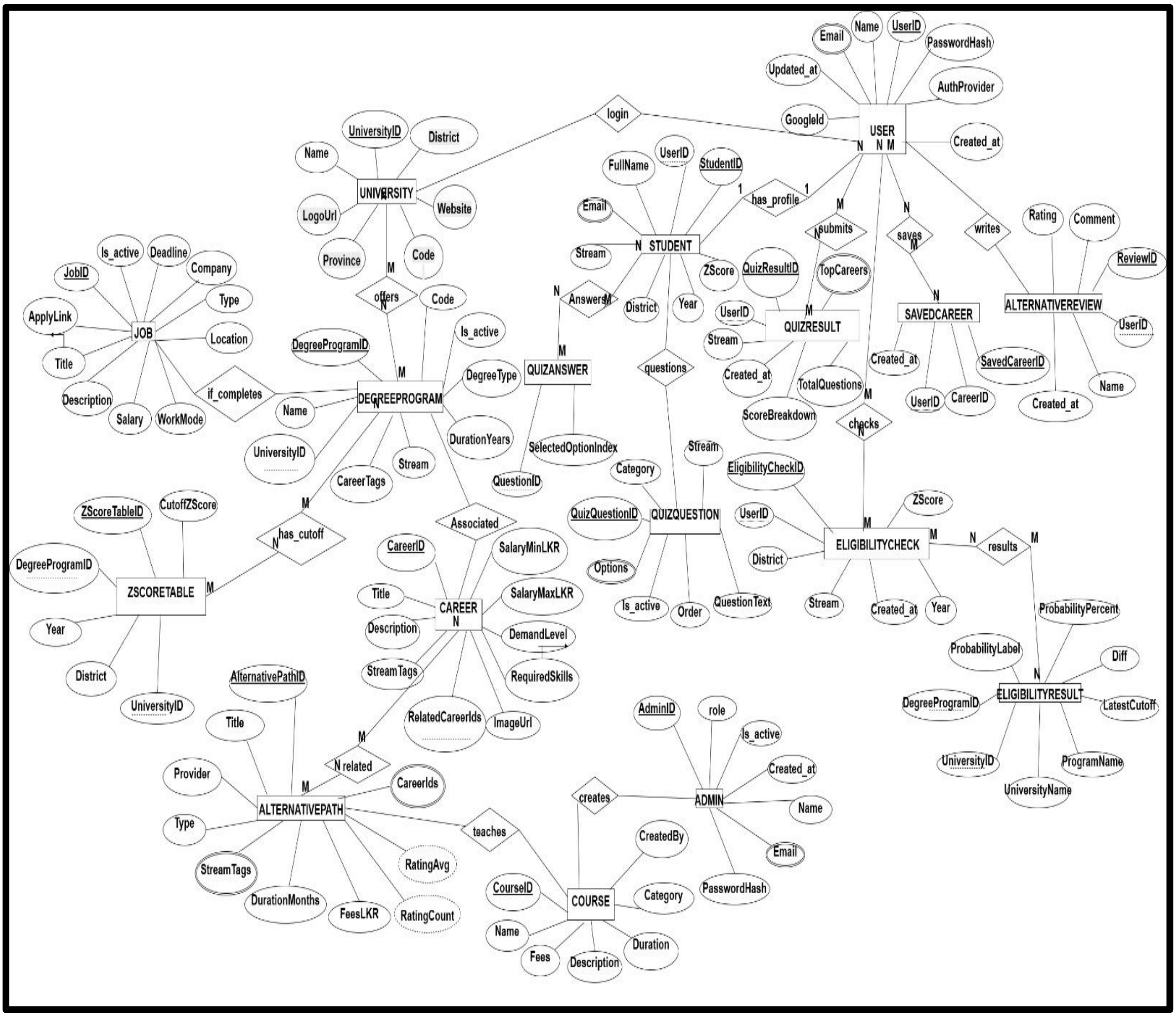

Figure 3.4MERN Career Platform Data Flow Diagram – showing data flows between Student, Professional, Admin, MERN Backend, MongoDB Database, Recommendation Engine, and Analytics Dashboard

3.5 Development Approach

The project followed an Agile-inspired iterative development approach with weekly sprint cycles aligned to the project timeline. Development was conducted using a feature-branch Git workflow, with each team member working on their assigned module in a dedicated branch before merging to the main repository via pull requests.

The project repository at GitHub tracked all commits, pull requests, and issue resolutions. By Week 4 of development, the team had made 23 commits to the main branch from 4 active contributors, with 4 open and 1 closed pull requests. Regular team meetings were held to review progress, resolve integration issues, and plan upcoming sprint work.

The system was developed and tested locally before integration testing was conducted on the combined application. Unit tests were written for core backend services including authentication, eligibility calculation, and quiz scoring logic.

Chapter 4: Results and Evaluation

4.1 System Outcomes

The completed system successfully implements all five core modules as defined in the project objectives. The following describes the key outcomes for each module as evidenced by the delivered user interfaces.

4.1.1 Authentication and User Management

The system provides a user authentication page supporting email/password login and Google OAuth. Upon successful authentication, users are directed to a personalized dashboard that provides quick-access links to all core system functions including Explore Careers, View Courses, Take a Quiz, Check Eligibility, Career Pathways, and Saved Careers. The dashboard displays the user's account email, session status, and authentication provider.

4.1.2 Career Assessment Module

The Career Assessment page allows students to select their academic stream or choose All Streams before initiating the assessment. The quiz interface presents 20 questions with four answer options each and displays a progress bar showing completion percentage. Upon completion, the results dashboard displays a score breakdown across career categories (Business, Education, Medicine, Design, IT, Engineering, Law) and presents the top three career matches with descriptions and score values. In testing with sample users, the scoring logic correctly identified the highest-scoring category based on answer patterns.

4.1.3 Qualification and Eligibility Analyzer

The eligibility analyzer interface allows students to input their full name, email, stream, district, Z-score, and examination year. The results page displays eligible degree programmes with probability labels, the student's Z-score versus the programme cut-off, the score difference, and historical cut-off values for 2021, 2022, and 2023. In a test case with stream Bio, district Jaffna, and Z-score 2.40 for year 2023, the system correctly identified five eligible programmes including BSc Agriculture (UOP), BSc Biological Science (UOK), and BSc Pharmacy (USJ) with High probability labels. The trend graph visualizes the three-year cut-off trajectory alongside the student's Z-score.

4.1.4 Career Path Explorer

The career listing page displays career cards with images, titles, salary ranges, demand levels, and stream tags. The filter panel allows filtering by stream, salary range, and demand level. In testing, filtering by salary range 100,000–300,000 LKR and demand level Medium correctly returned careers including Quantity Surveyor, Human Resources Manager, and DevOps Engineer. The detailed career page for Quantity Surveyor correctly displays the salary range (LKR 70,000–320,000), work environment, education paths, and required skills.

4.1.5 Alternative Pathway Recommender

The pathways page lists alternative options with ratings, provider names, duration, fees, and stream tags. The recommendation interface accepts student profile inputs and generates a ranked list of up to 10 recommendations scored out of 100. In a test case with A/L stream Maths, district Colombo, budget LKR 300,000, duration 24 months, government university eligibility No, and career interests Software and Engineering, the system returned BSc (Hons) in Information Technology from SLIIT as the top recommendation with a score of 100/100. The comparison page (UI 15) allows side-by-side evaluation of selected pathways across provider, type, duration, fees, rating, streams, career tags, districts, and requirements.

4.1.6 Admin and Job Management System

The admin dashboard provides summary statistics showing 3 users, 19 careers, 10 universities, 20 programmes, 20 quiz questions, and 3 jobs. Recent user and recent job panels display the latest activity. The Jobs and Internships page lists available positions with type, location, salary, and posting date. The detailed job view displays full job information including work mode, deadline, and application status.

4.2 Testing Results

Functional testing was conducted across all five modules using structured test cases. Each test case verified that system outputs matched expected results for defined inputs.

Table 4.2: Functional Test Results Summary

TC	Test Description	Expected Result	Actual Result	Status
TC01	User registration with valid email and password	Account created, redirect to dashboard	Account created successfully	Pass
TC02	Google OAuth login	User authenticated via Google, dashboard loaded	OAuth login successful	Pass
TC03	Career quiz – 20 questions answered	Score breakdown and top 3 careers displayed	Results displayed correctly	Pass
TC04	Eligibility check – Bio, Jaffna, Z=2.40, 2023	5 eligible programmes displayed with High label	5 programmes shown with correct labels	Pass
TC05	Career filter – Salary 100K–300K, Medium demand	Filtered careers displayed	Correct careers returned	Pass
TC06	Pathway recommendation – Maths, Software, LKR 300K	SLIIT BSc IT ranked #1 with score 100/100	BSc IT ranked first as expected	Pass
TC07	Admin – Add new job listing	Job appears on public jobs page	Job listed correctly	Pass
TC08	Trend graph for BSc Pharmacy	3-year cut-off trend visualized with student Z-score line	Graph rendered correctly	Pass
TC09	Pathway comparison – 2 pathways	Side-by-side table with all attributes	Comparison table displayed accurately	Pass
TC10	Admin dashboard – statistics summary	Correct counts for users, careers, jobs	All counts accurate	Pass

4.3 Performance Metrics

Basic performance testing was conducted by measuring page load times under standard network conditions. The home page loaded in approximately 1.2 seconds, the career listing page in 1.8 seconds, and the eligibility results page in 2.1 seconds. All measured load times were within the three-second performance requirement defined in NFR1.

The system was also tested with concurrent access by five users simultaneously performing different operations including quiz completion, eligibility checking, and career browsing. No significant performance degradation was observed during these tests.

4.4 User Feedback

Informal user feedback was collected from a small group of O/L and A/L students who tested the system. The feedback was generally positive, with participants highlighting the clarity of the eligibility results, the usefulness of the trend graph for understanding cut-off patterns, and the ease of navigation through the career explorer. Suggestions for improvement included adding more careers to the database, expanding the quiz question bank, and providing a mobile application in the future. These suggestions have been noted for the system's next development phase.

Chapter 5: Conclusion

5.1 Summary of Achievements

This project has successfully designed, developed, and evaluated a Web-Based Career Selection and Education Guidance System tailored to the specific needs of Sri Lankan school leavers following their O/L and A/L examinations. The completed system delivers five fully functional modules the Career Assessment Module, the Qualification and Eligibility Analyzer, the Career Path Explorer, the Alternative Pathway Recommender, and the Admin and Job Management System integrated into a coherent, user-friendly web application.

5.2 Objectives Achievement

Each of the six project objectives defined in Chapter 1 has been met:

- A career exploration platform has been implemented that provides detailed information on careers including required qualifications, salary ranges, demand levels, and skills.
- The eligibility analyzer uses historical Z-score data to determine likely eligibility for government university programmes and displays results with probability labels and trend graphs.
- Personalized career recommendations are generated based on interest and skill assessment quiz responses.
- The alternative pathway recommender suggests private degrees, diplomas, vocational training, and foundation programmes for students who do not qualify for their preferred university courses.
- Side-by-side pathway and career comparison functionality has been implemented to support informed decision-making.
- A secure administrative dashboard has been developed to enable content managers to manage all system data.

5.3 Project Aim Achievement

The overall project aim to develop a web-based system that supports informed career and educational decision-making for Sri Lankan school leavers has been achieved. The system consolidates career guidance, eligibility analysis, and pathway recommendation into a single, accessible platform that is contextually relevant to the Sri Lankan educational system. Functional testing confirms that all core modules operate as intended, and initial user feedback indicates that the system provides meaningful guidance value to its target users.

Looking ahead, the system provides a strong foundation for future enhancements including AI-driven recommendation improvements, mobile application development, integration with UGC and ministry data sources, and expansion of the career and pathway databases. The project has also demonstrated the value of applying structured software engineering principles including requirement analysis, modular design, iterative development, and testing to the development of socially impactful digital tools.

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Appendix A: Team Member Contributions

This appendix presents the individual contributions of each team member to the project. All five members have reviewed and agreed on the contribution percentages listed below. Contributions are linked to verifiable work evidence including Git commits, diagrams produced, UI pages developed, and testing performed.

The project Git repository is available at:

https://github.com/IT24102099/A-Web-Based-Career-Selection-and-Education-Guidance-System

Name	Primary Responsibilities	Contribution (%)	Development Work	Diagrams / Design Work	Testing & Validation
Mariyappan Thamilpiriyan **IT23296800**	Developed Career Path Explorer including career listing page, filtering system, detailed career pages, save and compare functionality. Acted as project lead and managed Git repository and branches.	20%	Implemented career UI components, filtering logic, and comparison features	Designed Career Explorer related UI structures and contributed to feature planning	Conducted functional testing for filtering, career display, and comparison features
Kamshiga Ganesan **IT24101365**	Developed Alternative Pathway Recommender including recommendation engine, pathway listing, filtering, and comparison functionality	20%	Built recommendation logic, pathway UI pages, and filtering mechanisms	Designed pathway structure and recommendation flow	Tested recommendation scoring accuracy and pathway comparison functionality
Naguleswaran Mathuppriya	Developed Qualification & Eligibility Analyzer	20%	Implemented eligibility calculation logic, result	Created EER Diagram, Data Flow Diagram (DFD), and system	Performed testing on Z-score calculations, eligibility matching,

IT24102099	including eligibility form, results display, probability logic, and trend visualization. Also contributed to system design		display, and trend graph integration	architecture design	and trend graph accuracy
Sithirasenan Vipooshan **IT24102869**	Developed Career Assessment Module including quiz interface, scoring logic, and results dashboard	20%	Built quiz system with 20-question flow, scoring algorithm, and result generation	Contributed to Use Case Diagram and quiz design structure	Tested quiz flow, scoring correctness, and result accuracy
Moosika Ramanathan **IT24102839**	Developed Admin & Job Management System including admin dashboard, job listing, user management, and content control	20%	Implemented admin dashboard, CRUD operations, and job/internship management features	Designed admin panel structure and content management workflows	Conducted testing for admin CRUD operations, job listing accuracy, and dashboard data

The developed system has been successfully deployed using modern cloud hosting platforms to ensure accessibility and real-world usability.

- The frontend application (React) was deployed using Vercel, providing fast and reliable hosting with continuous deployment support.
- The backend server (Node.js/Express) was deployed using Render, enabling API hosting and database connectivity in a production environment.

These deployments allow the system to function as a fully operational web application accessible via the internet, demonstrating practical implementation beyond local development.

Deployment Links:

- **Frontend Application:** https://career-guidance-mern.vercel.app/
- **Backend Service (API):** https://career-guidance-mern.onrender.com

Appendix B: System User Interface Screenshots

This appendix contains screenshots of the main user interface pages of the completed Web-Based Career Selection and Education Guidance System. Screenshots were captured from the working prototype.

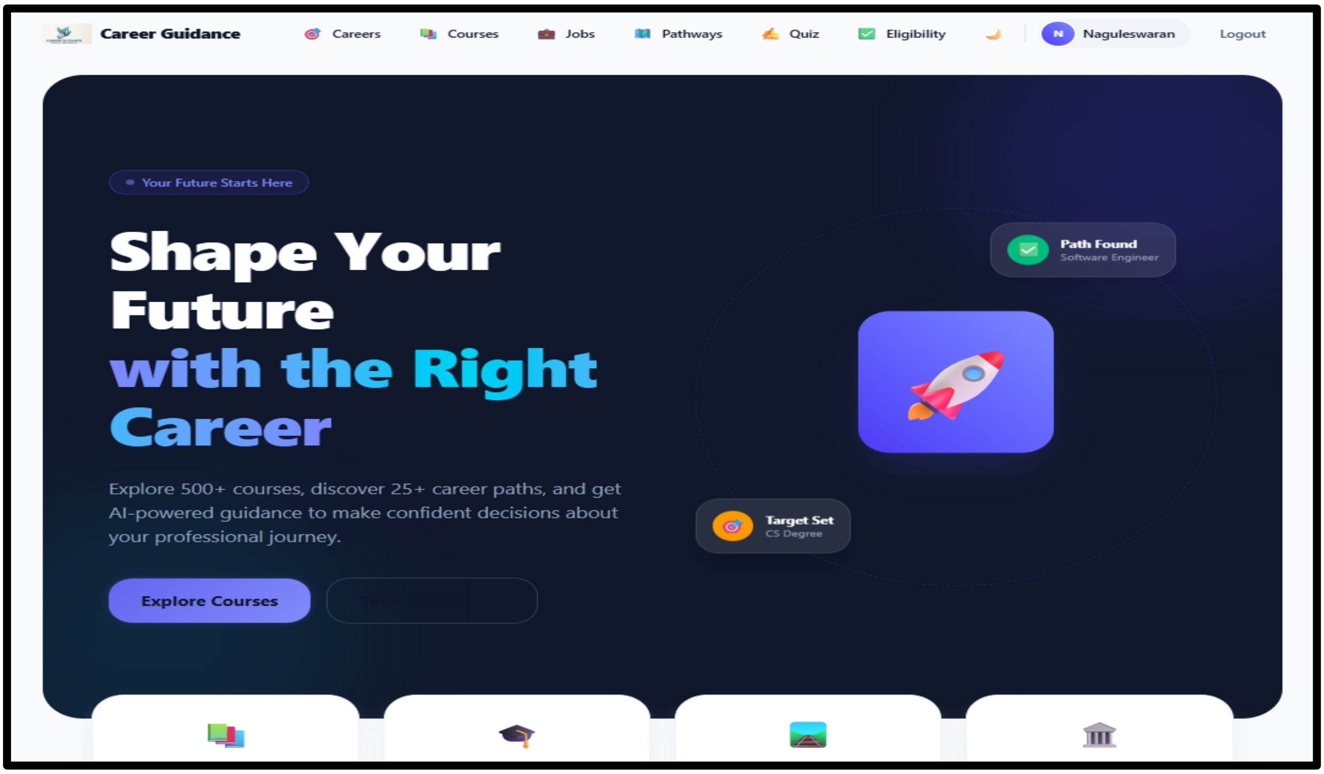


Figure B.1: The welcome Interface

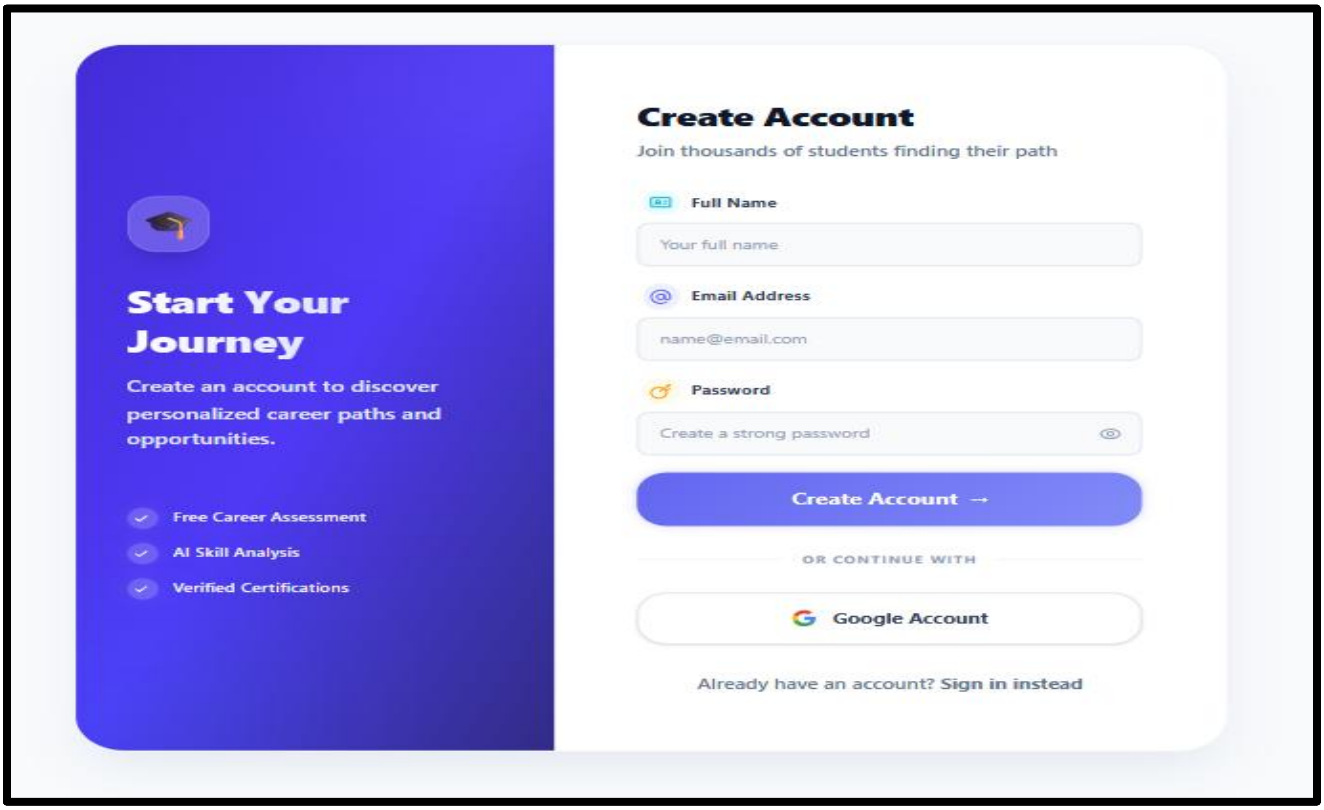


Figure B.2: User Login Interface

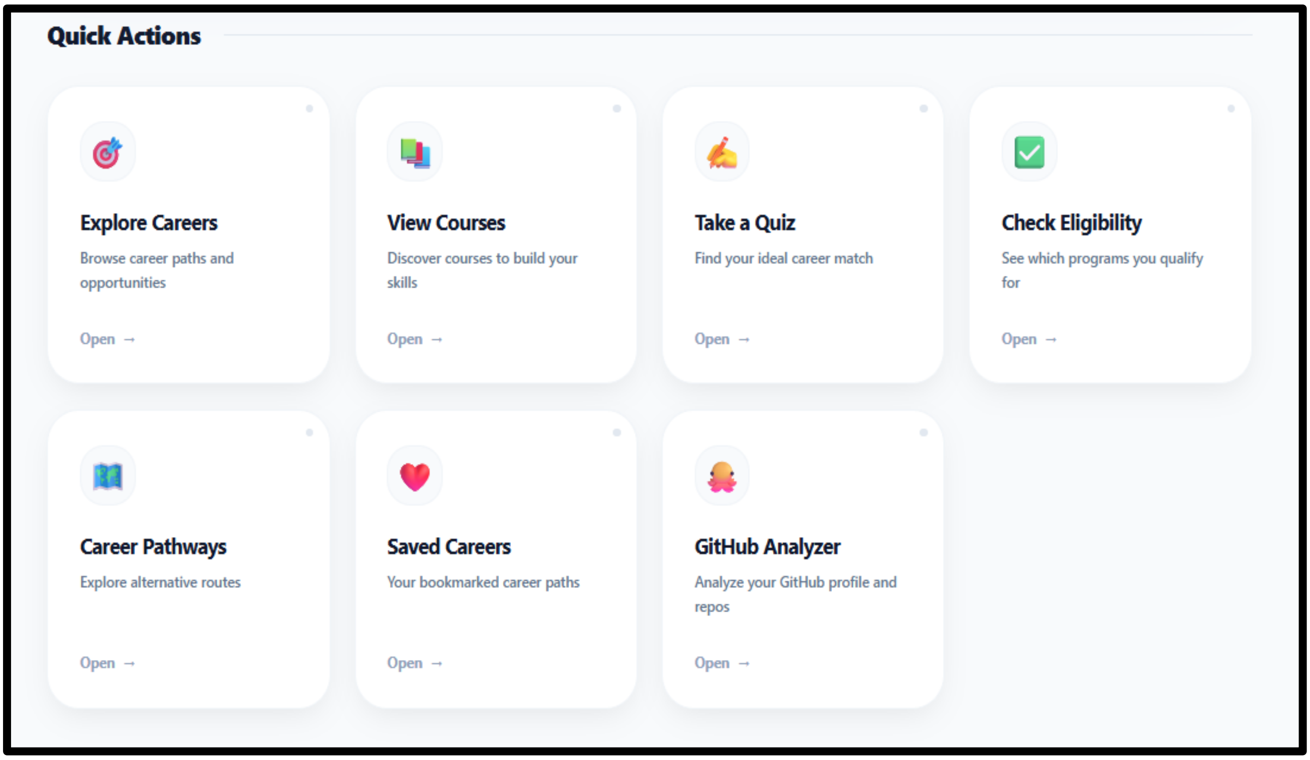


Figure B.3: Quick actions interface

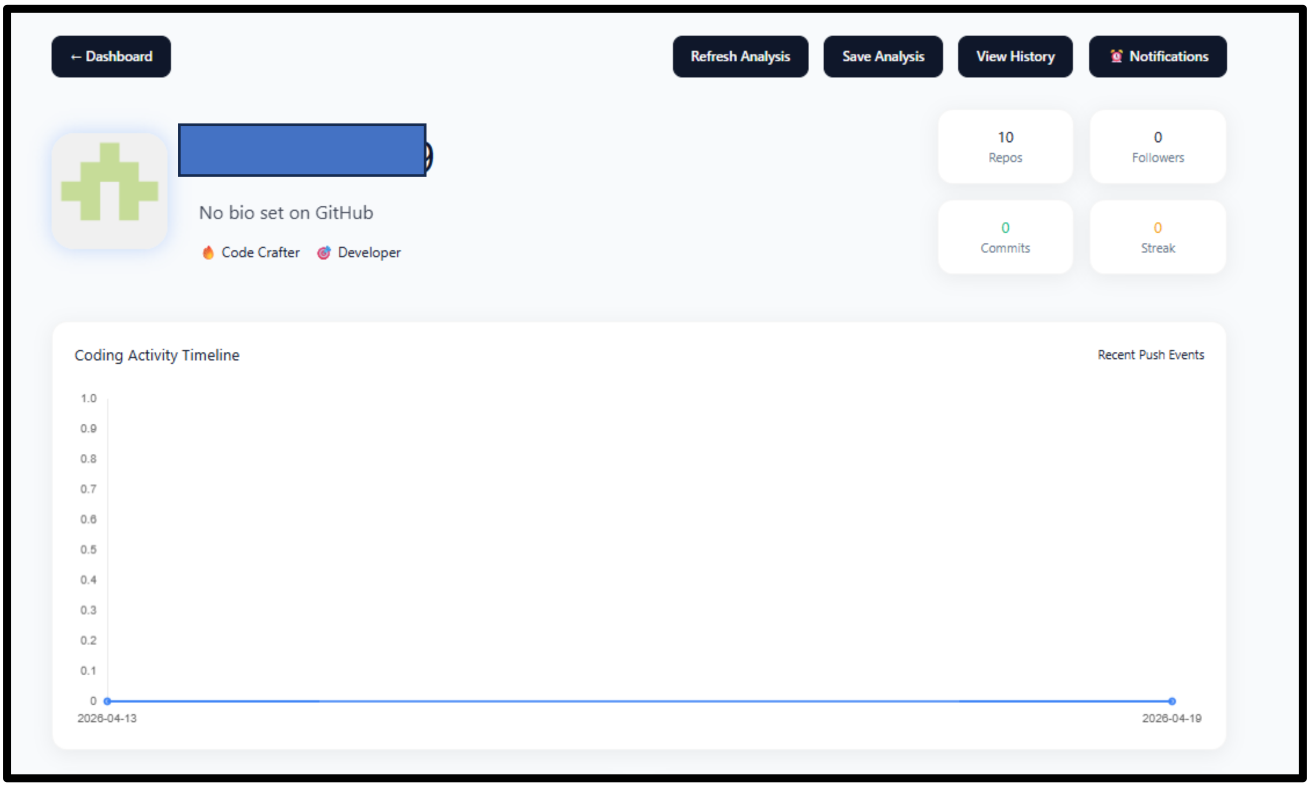


Figure B.4: GitHub analyzer interface

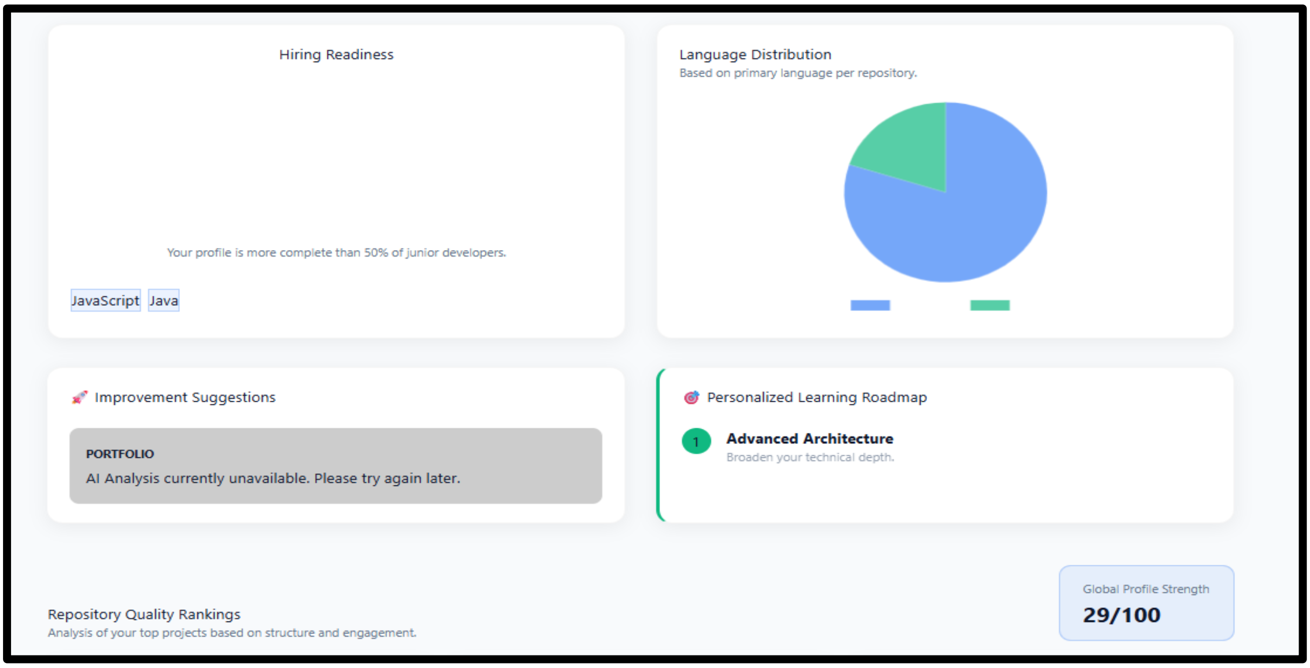


Figure B.5: Visual analyzing the profile.

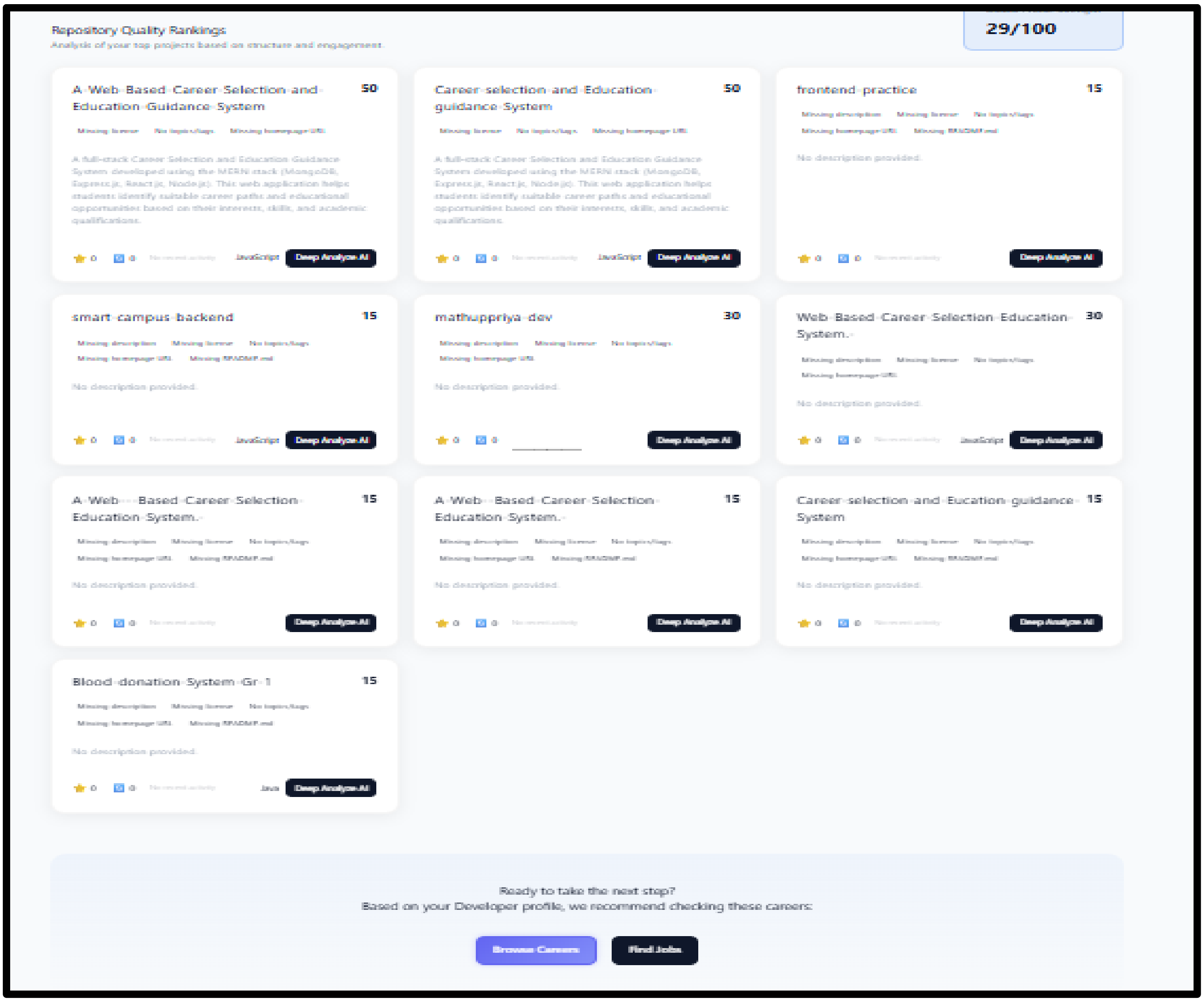


Figure B.2: Analyzing each repo in the account.

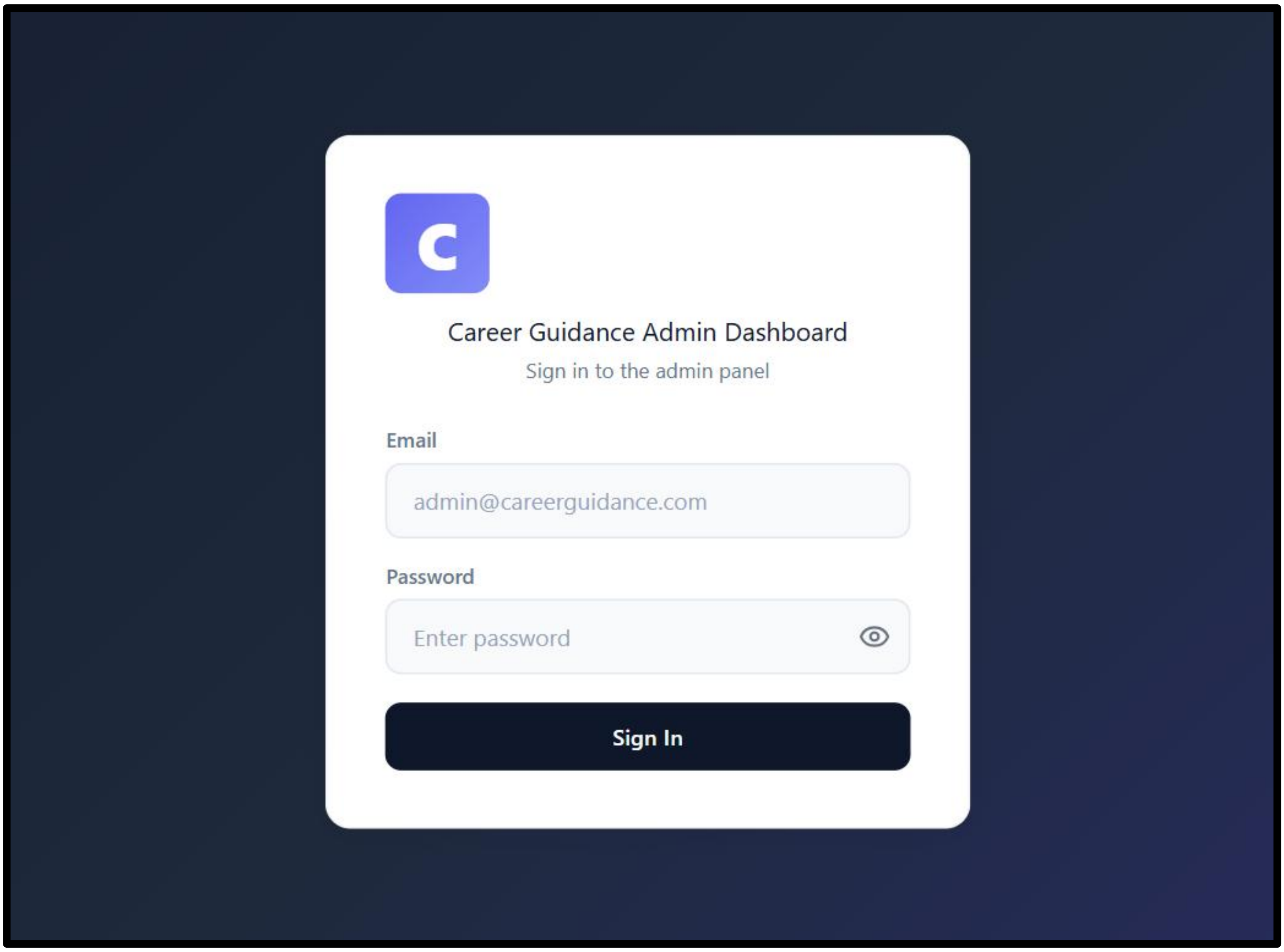


Figure B.3 : Admin Login Interface

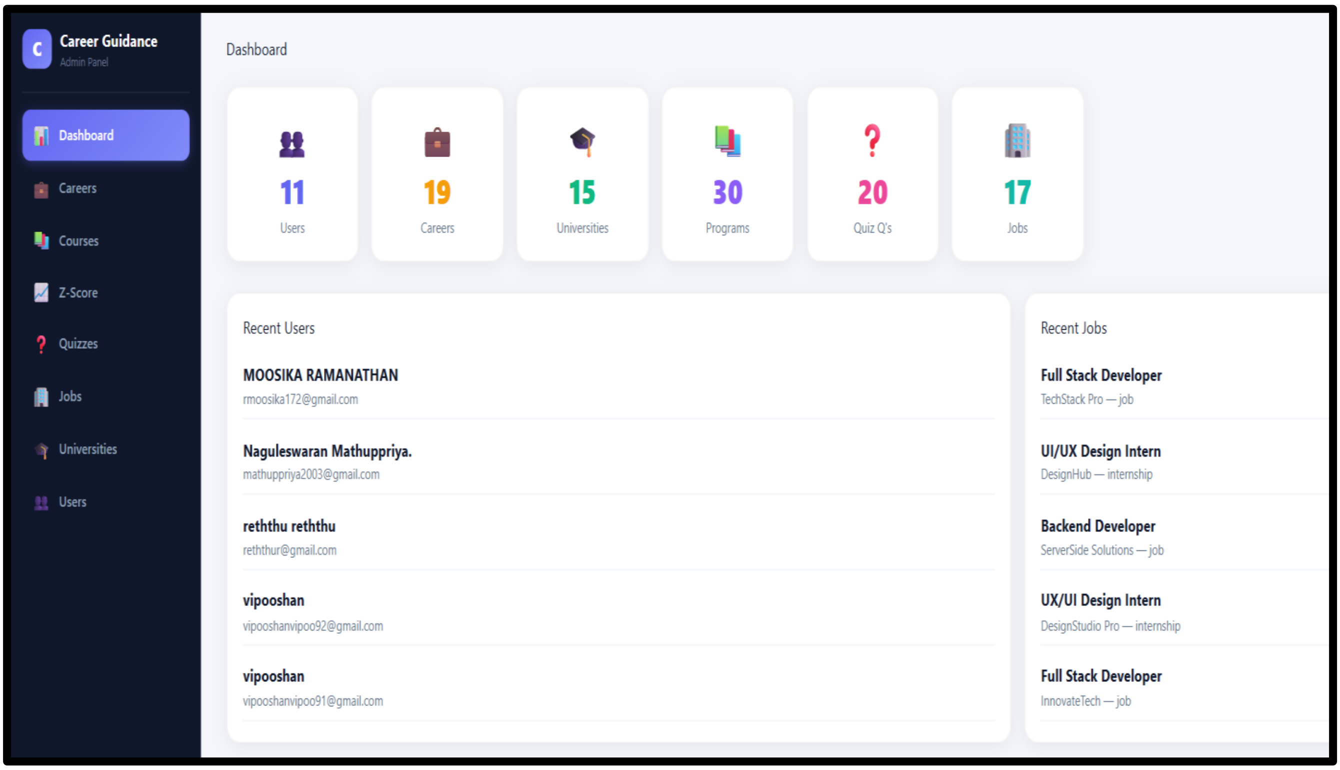


Figure B.4 : Admin Dashboard

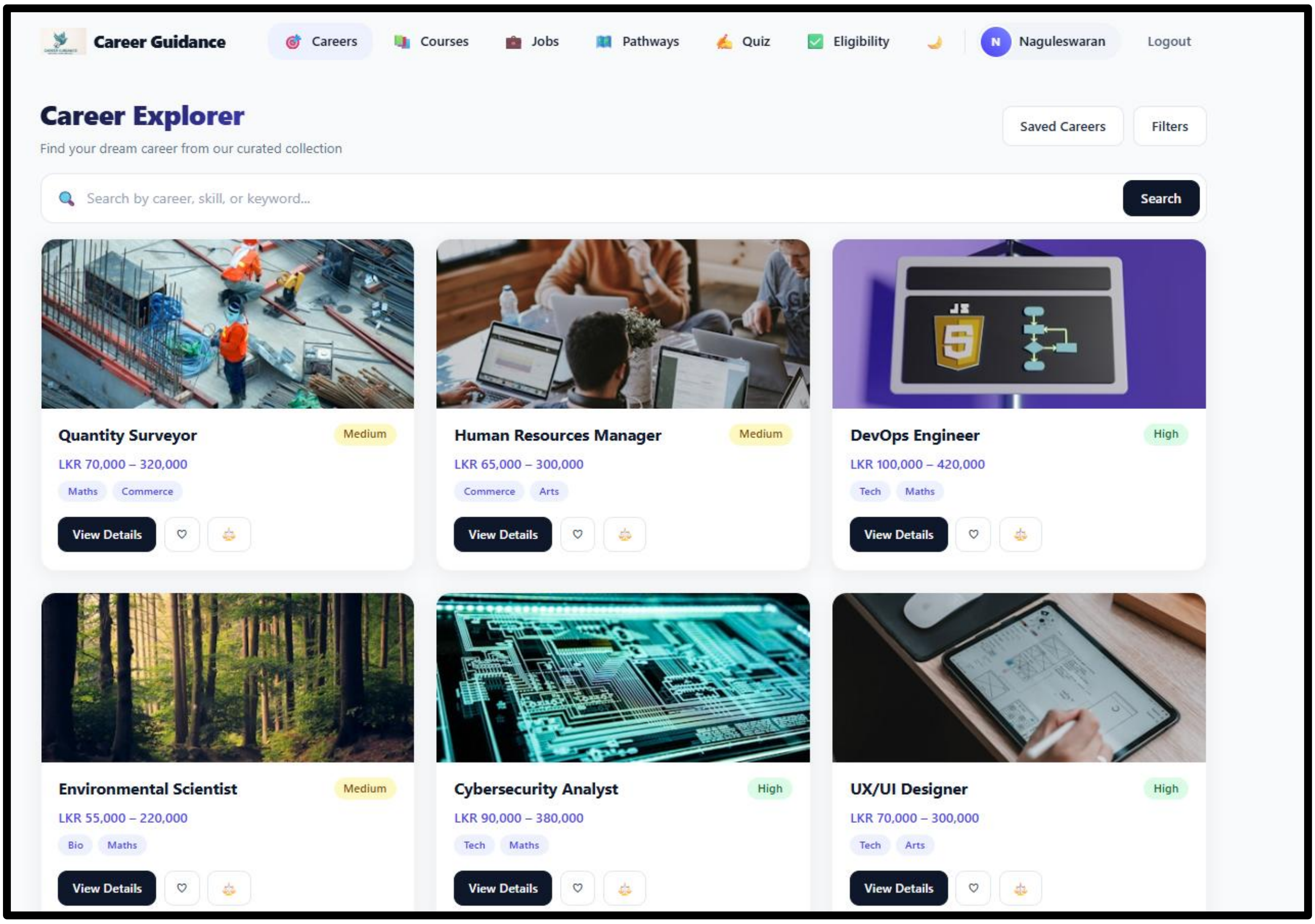


Figure B.5: Career Explorer Interface

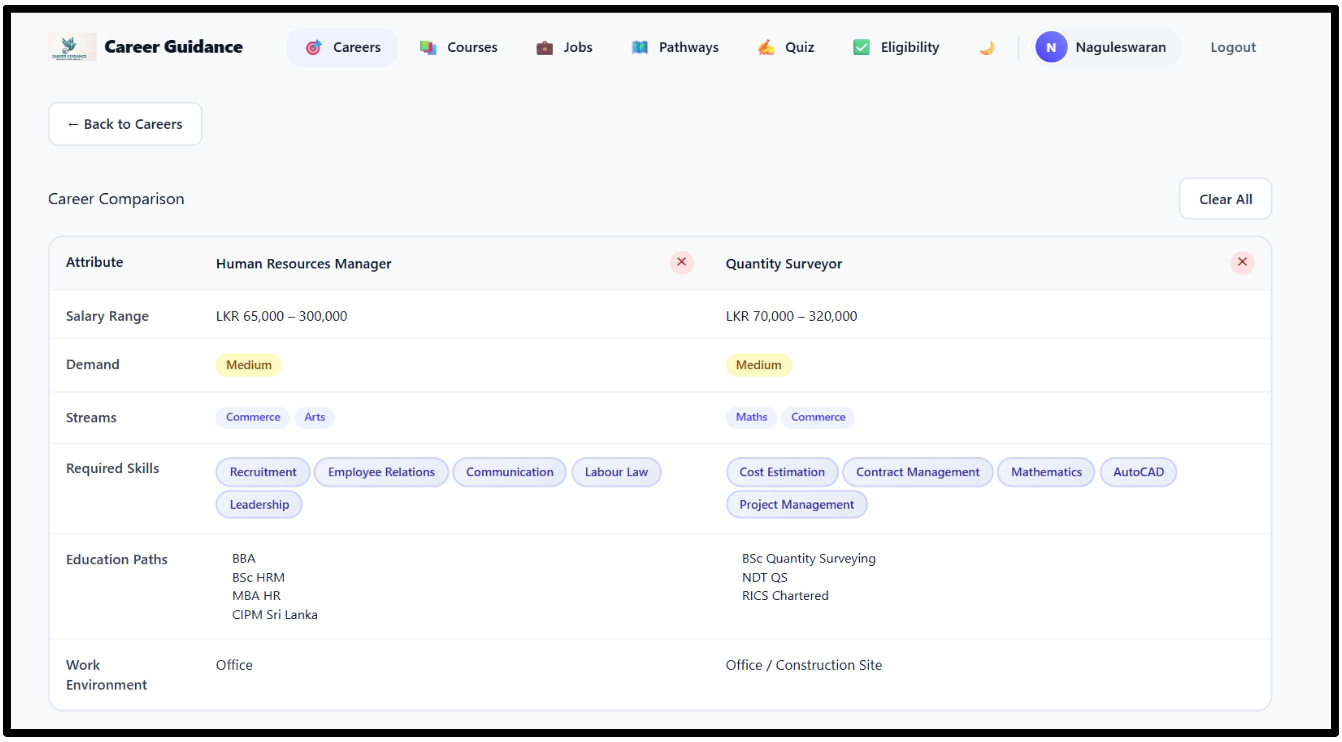


Figure B.6: Compare the careers

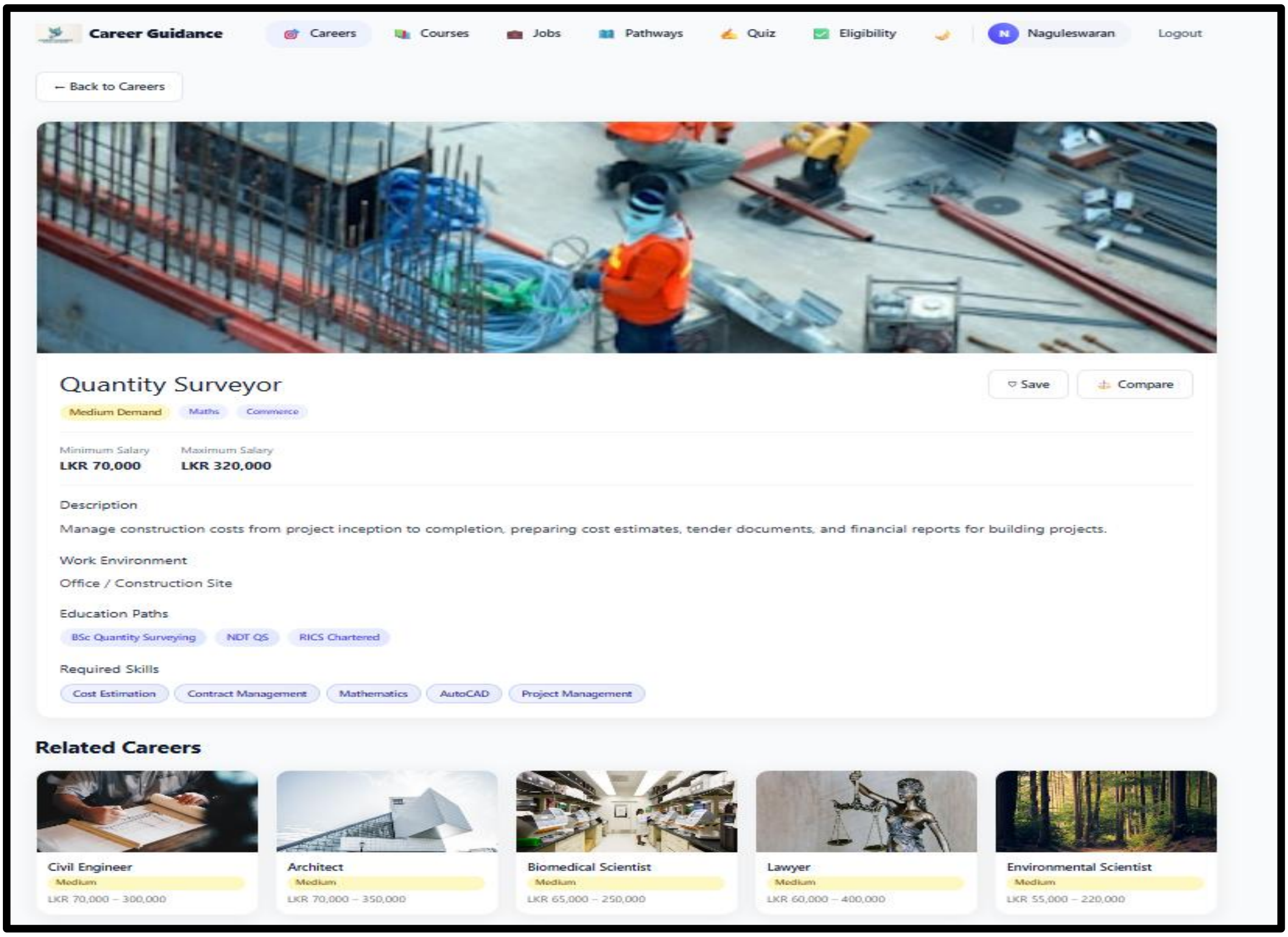


Figure B.6: Can see the details of each career.

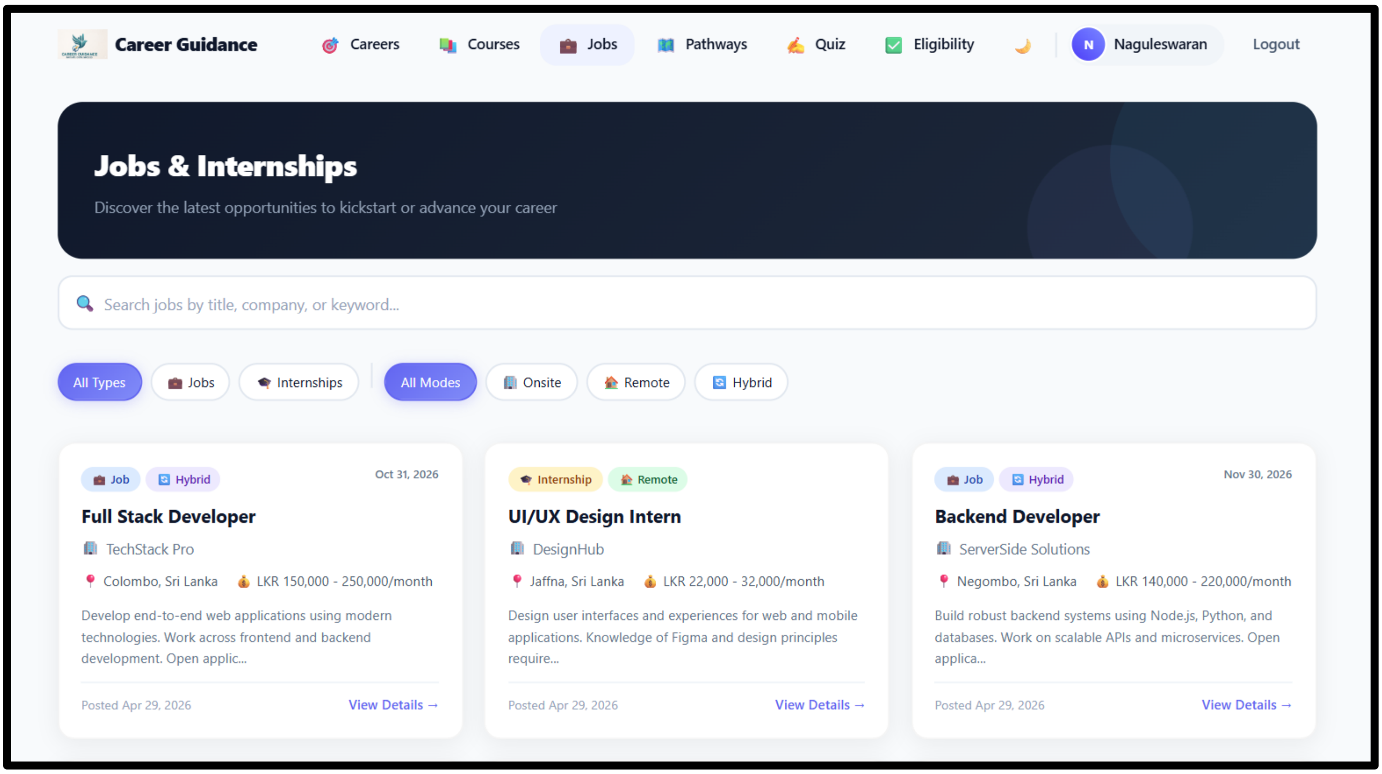


Figure B.6: Jobs & Internship Dashboard

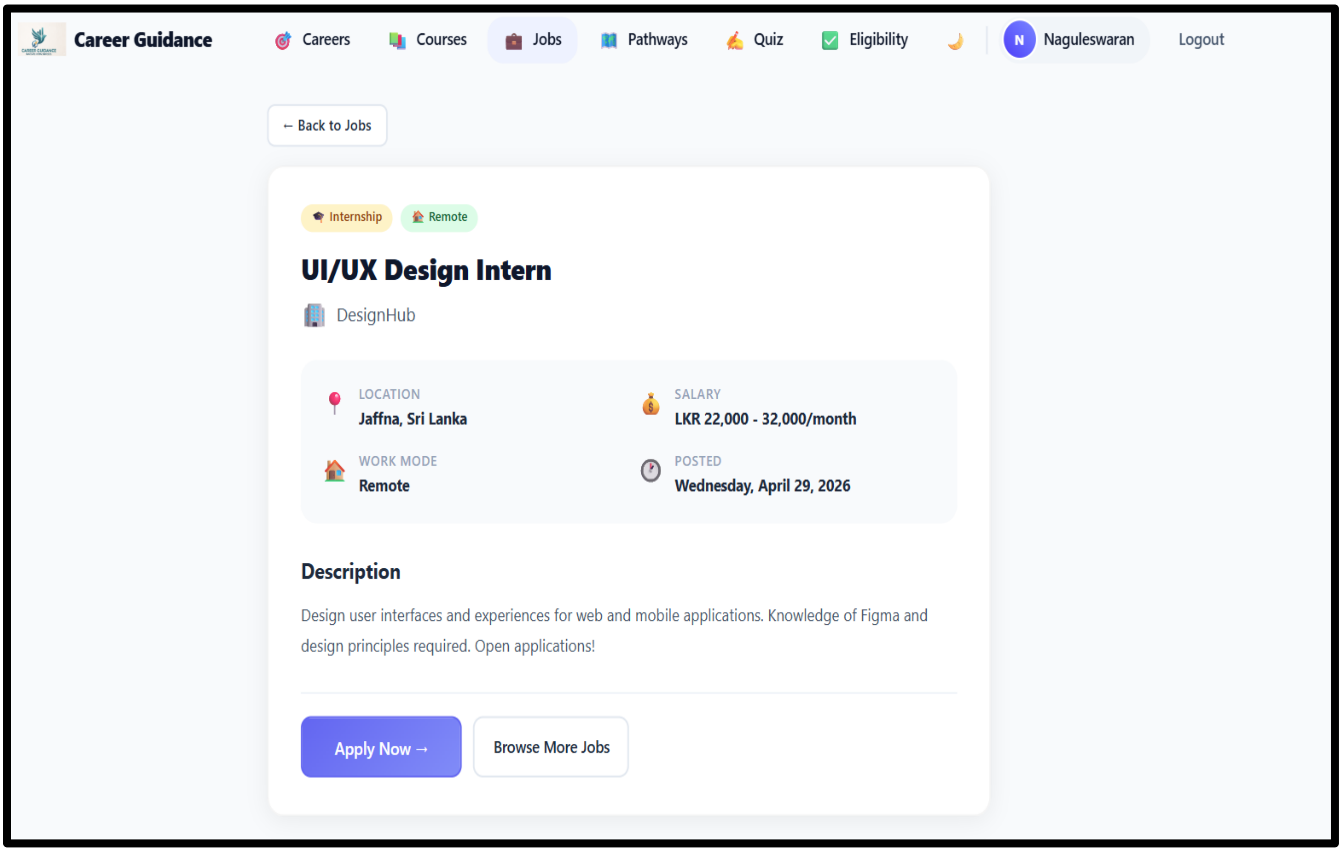


Figure B.6: Detail of the each Job/Intern

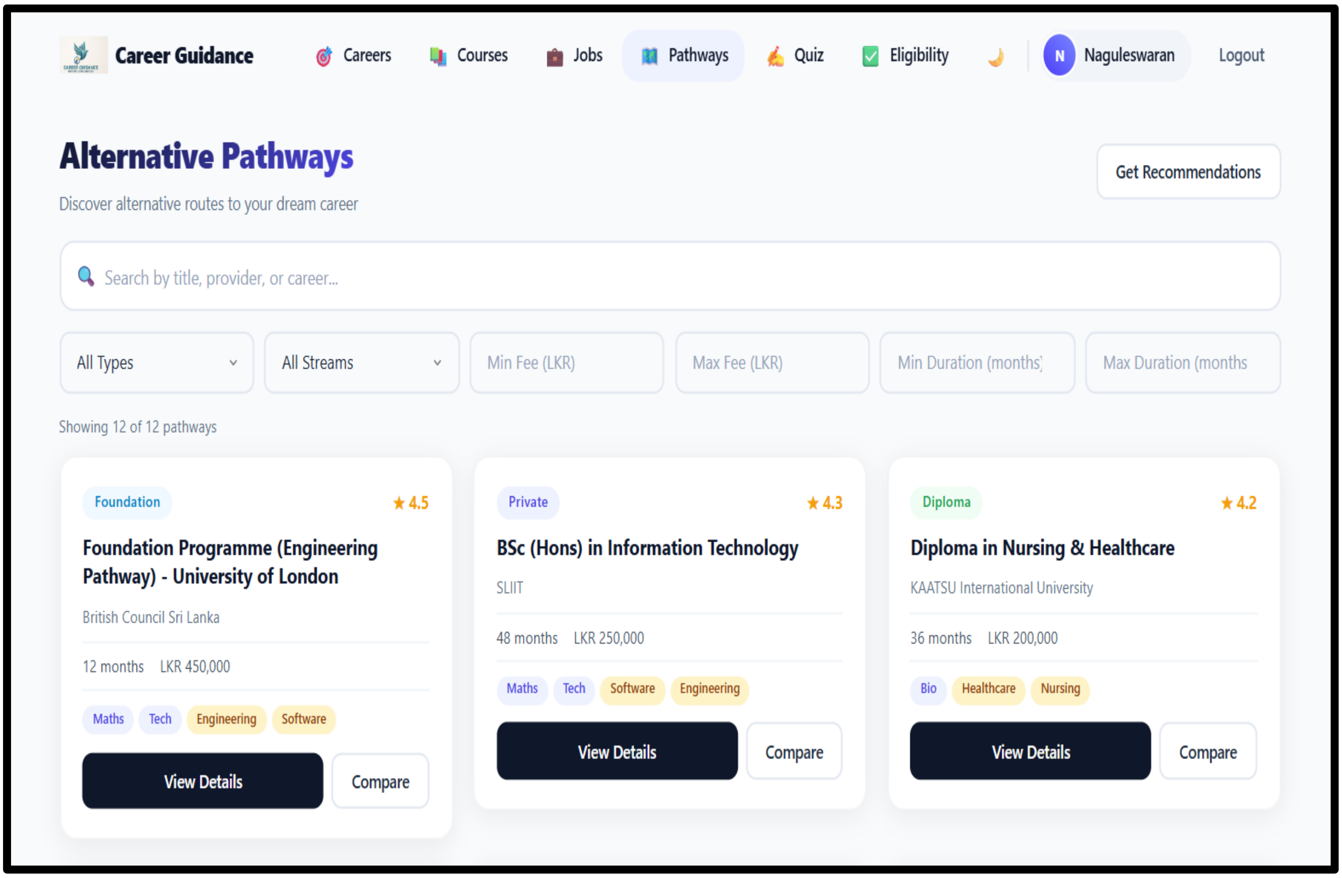


Figure B.6: Alternative pathway dashboard

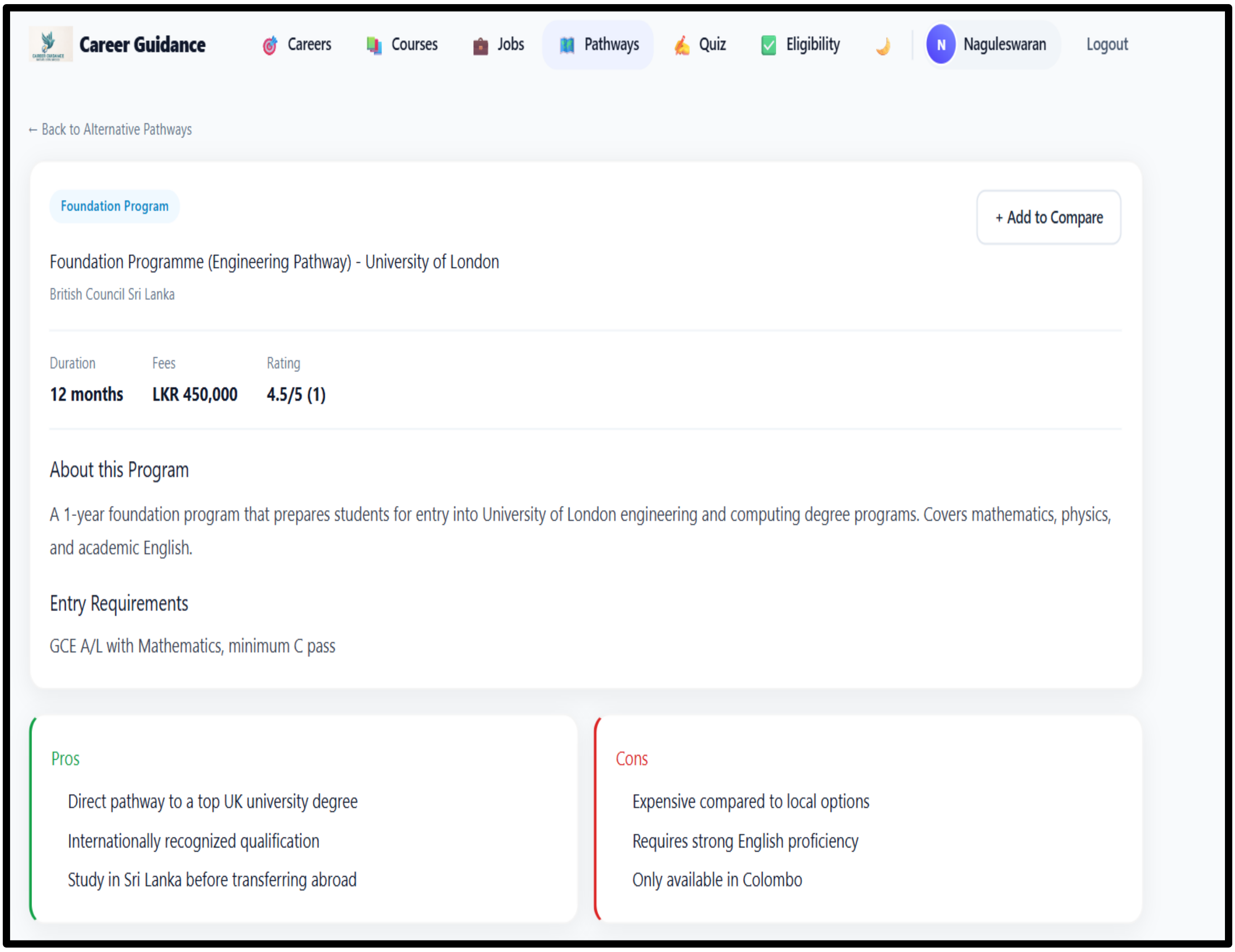


Figure B.6: Can view the complete detail of the programs.

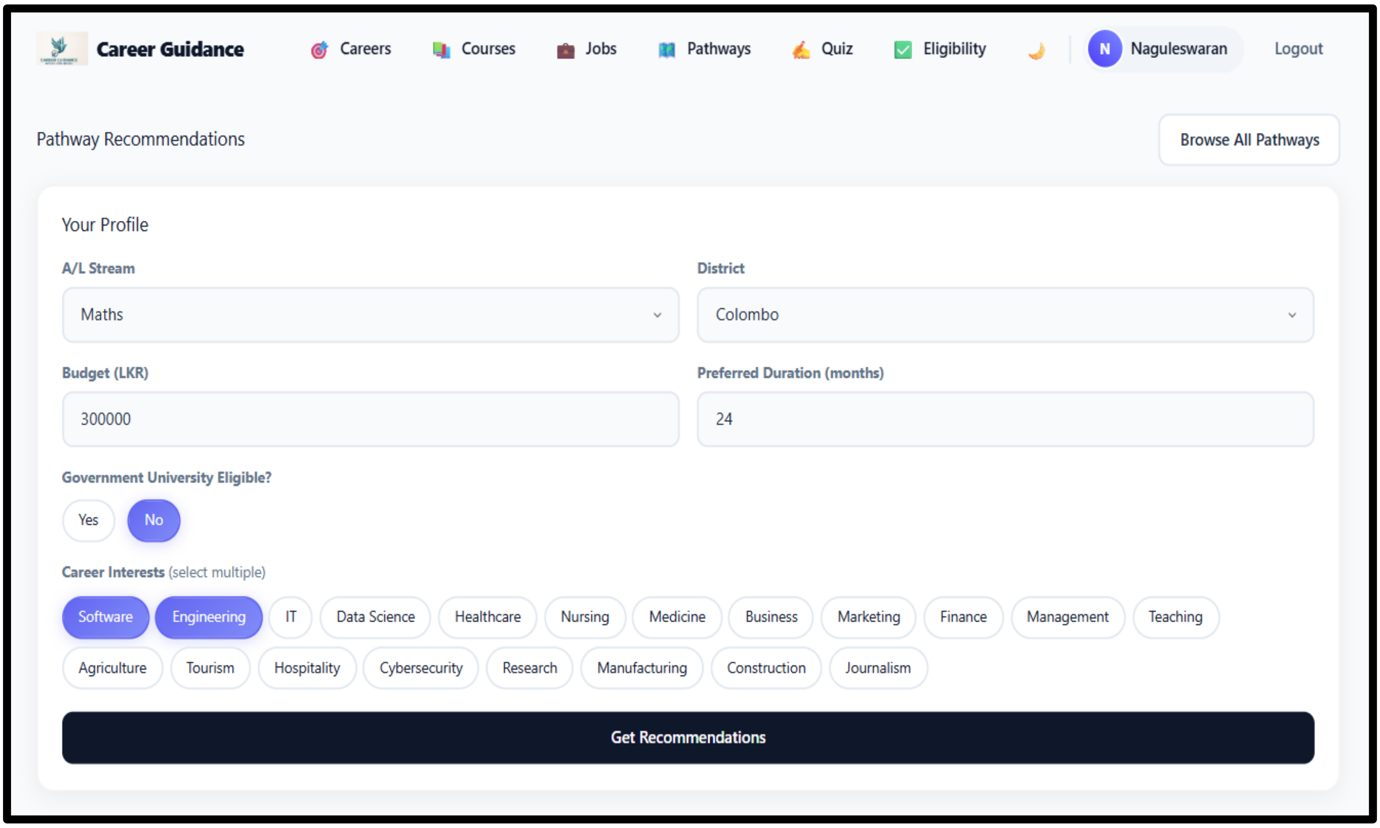


Figure B.6: Recommandation dashboard

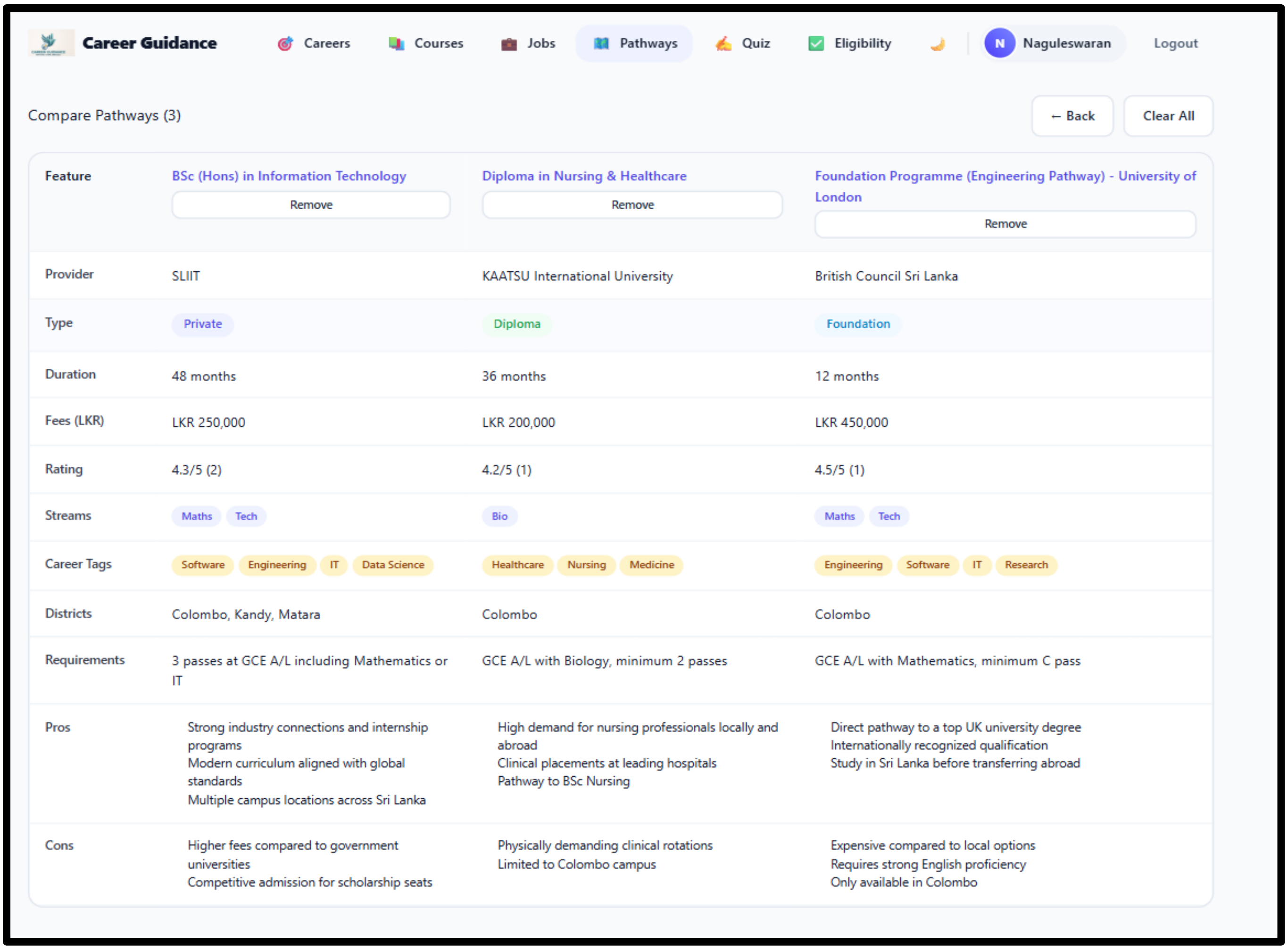


Figure B.6: Compare the programs

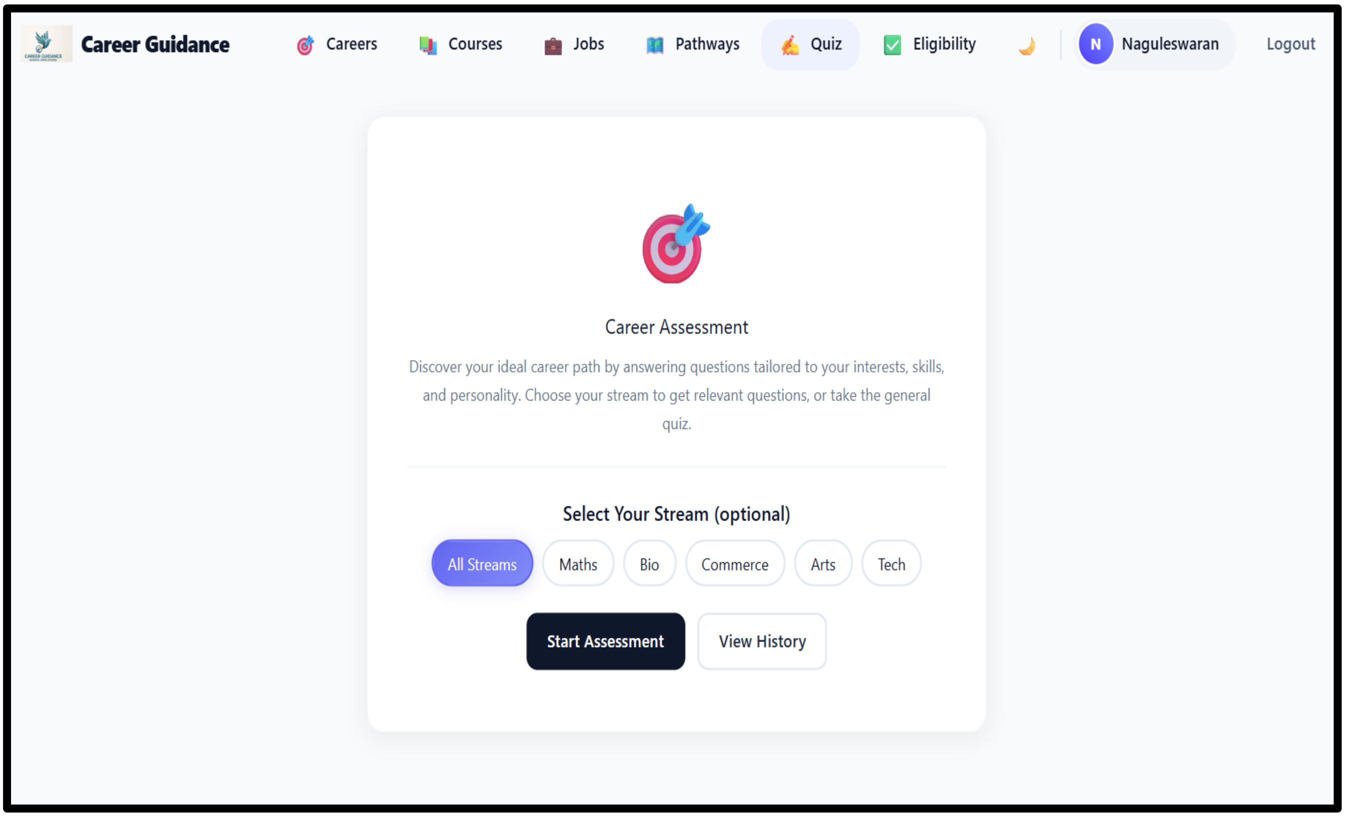


Figure B.6: Career assessment dashboard

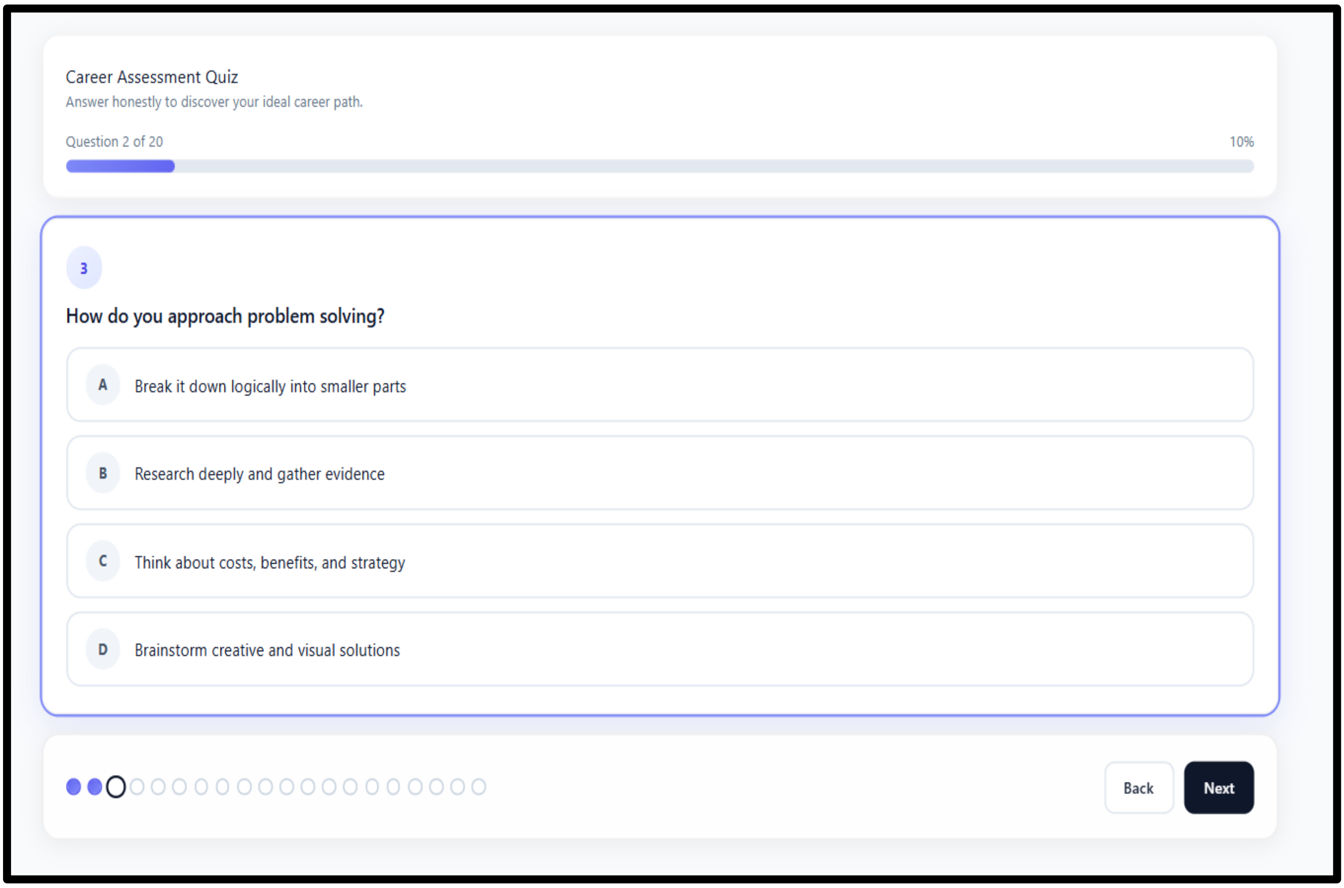


Figure B.6: Quiz page for the career assessment.

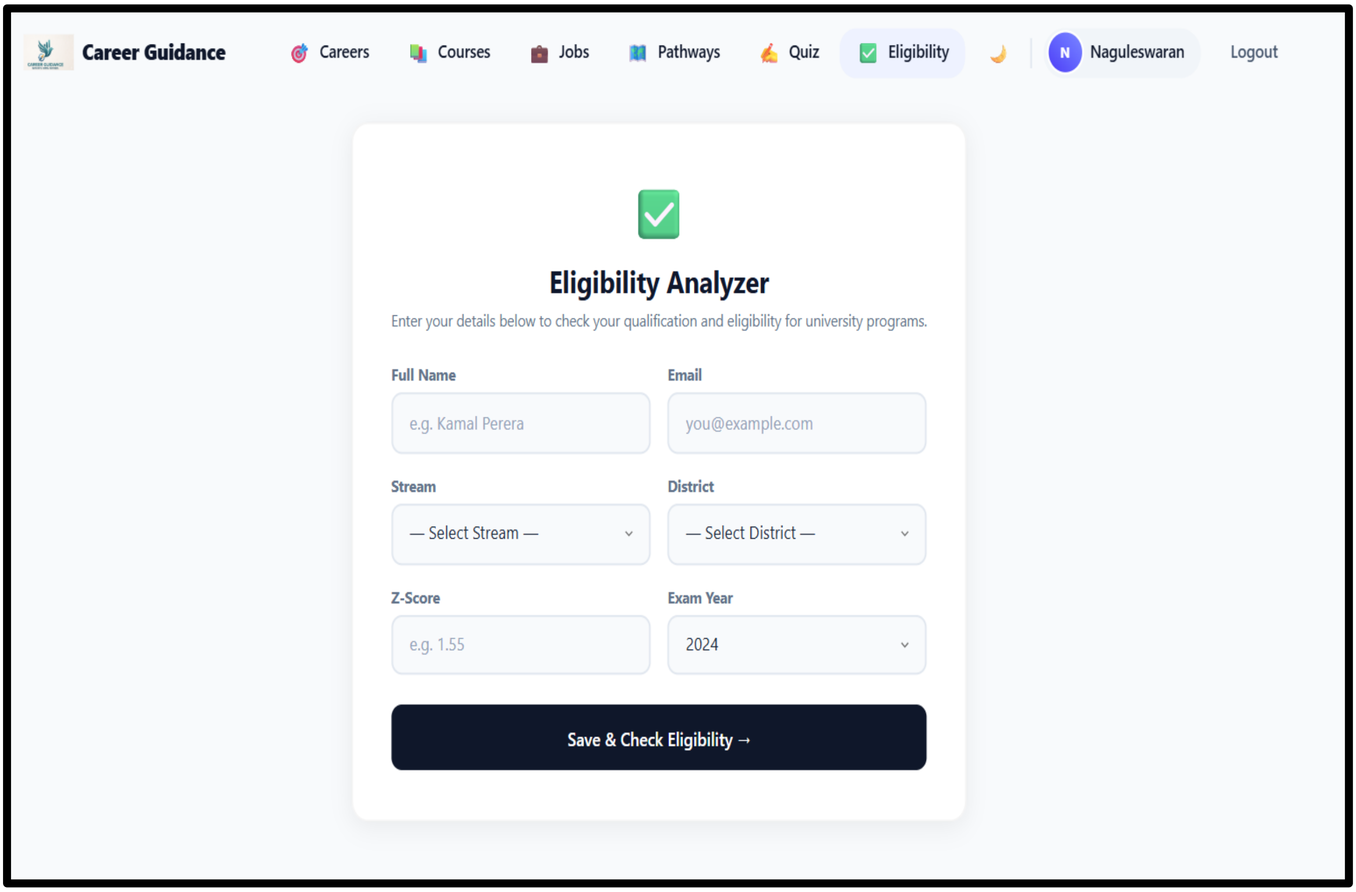


Figure B.6: Eligibility analysis form

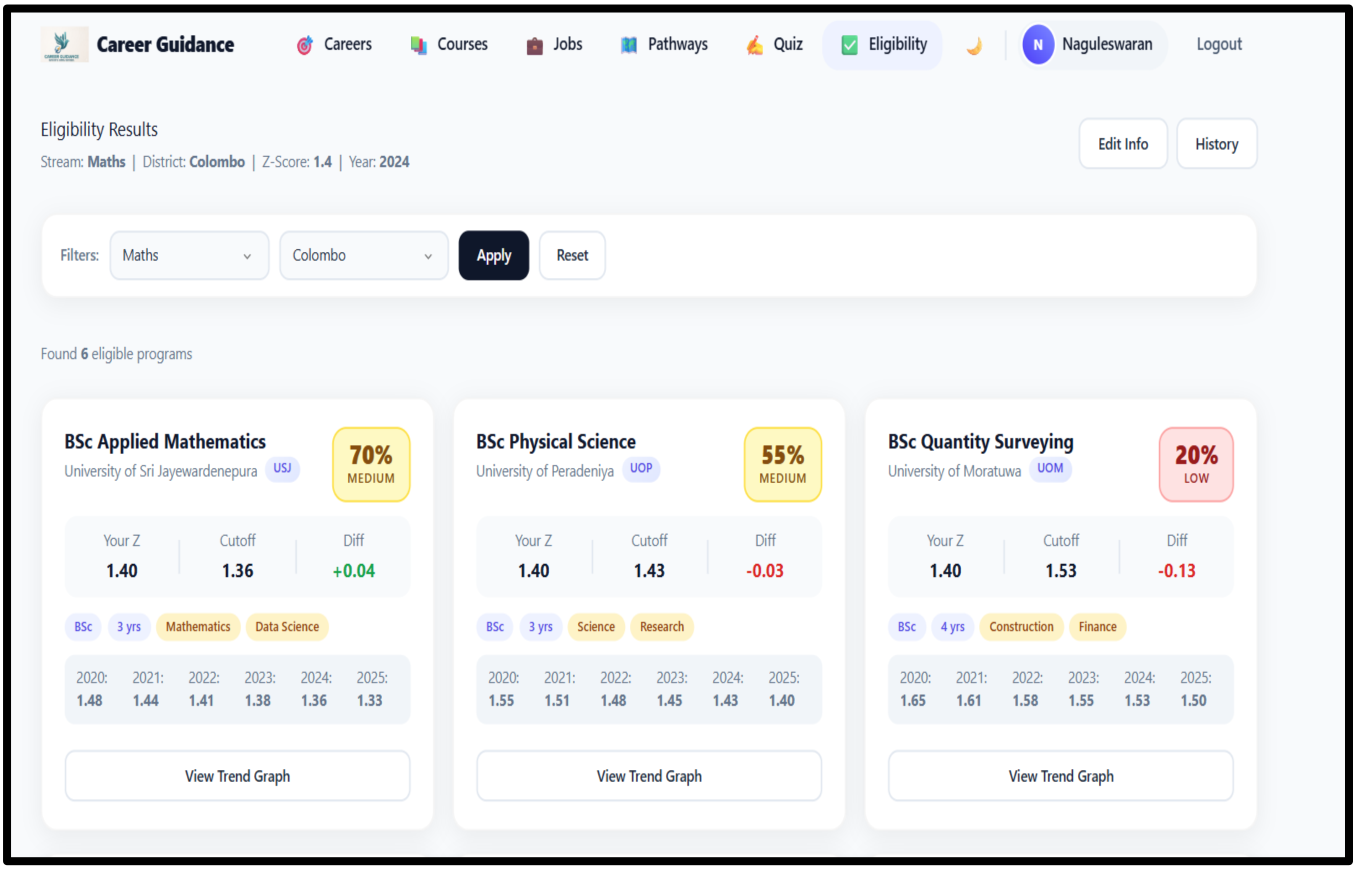
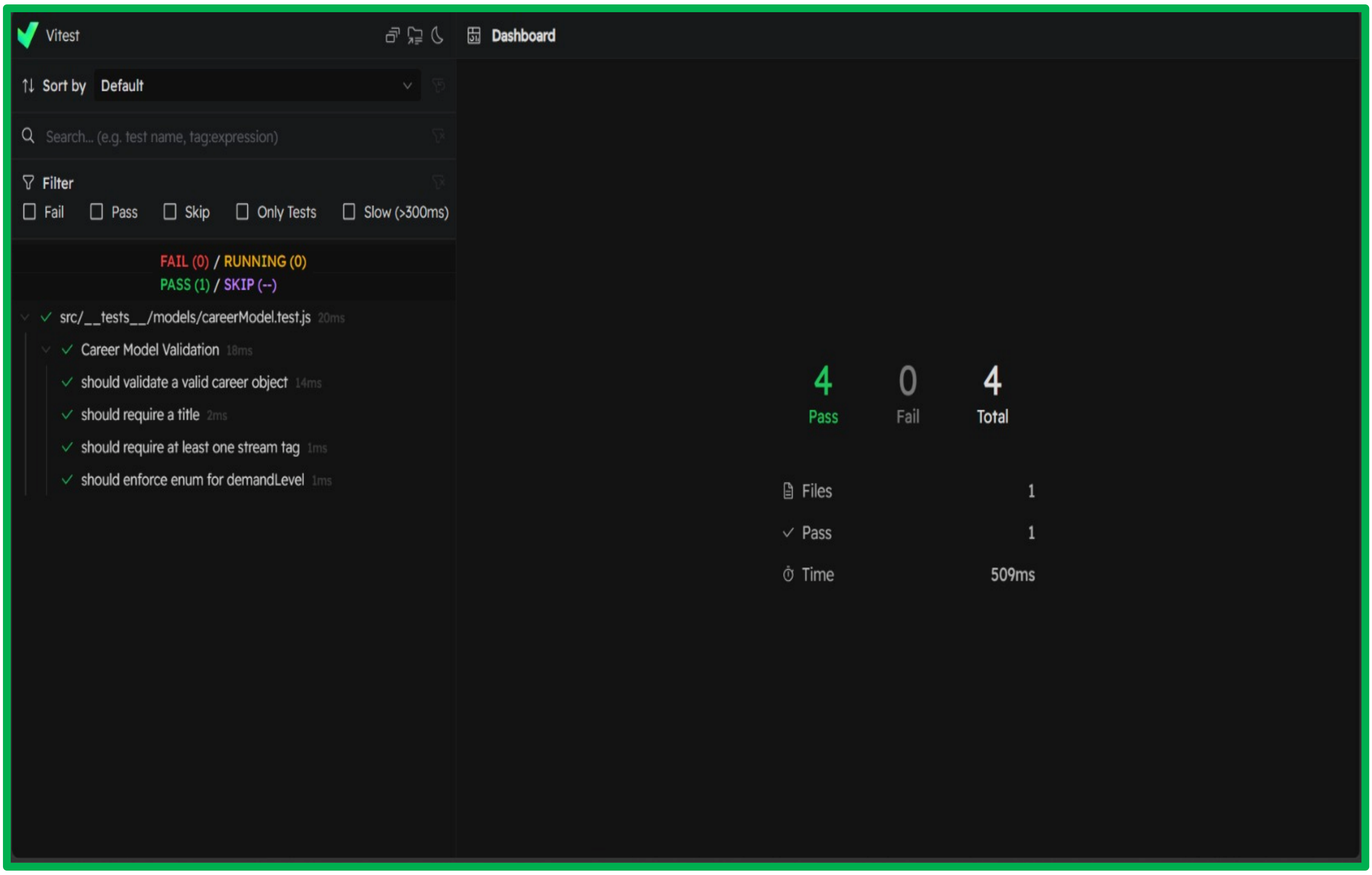
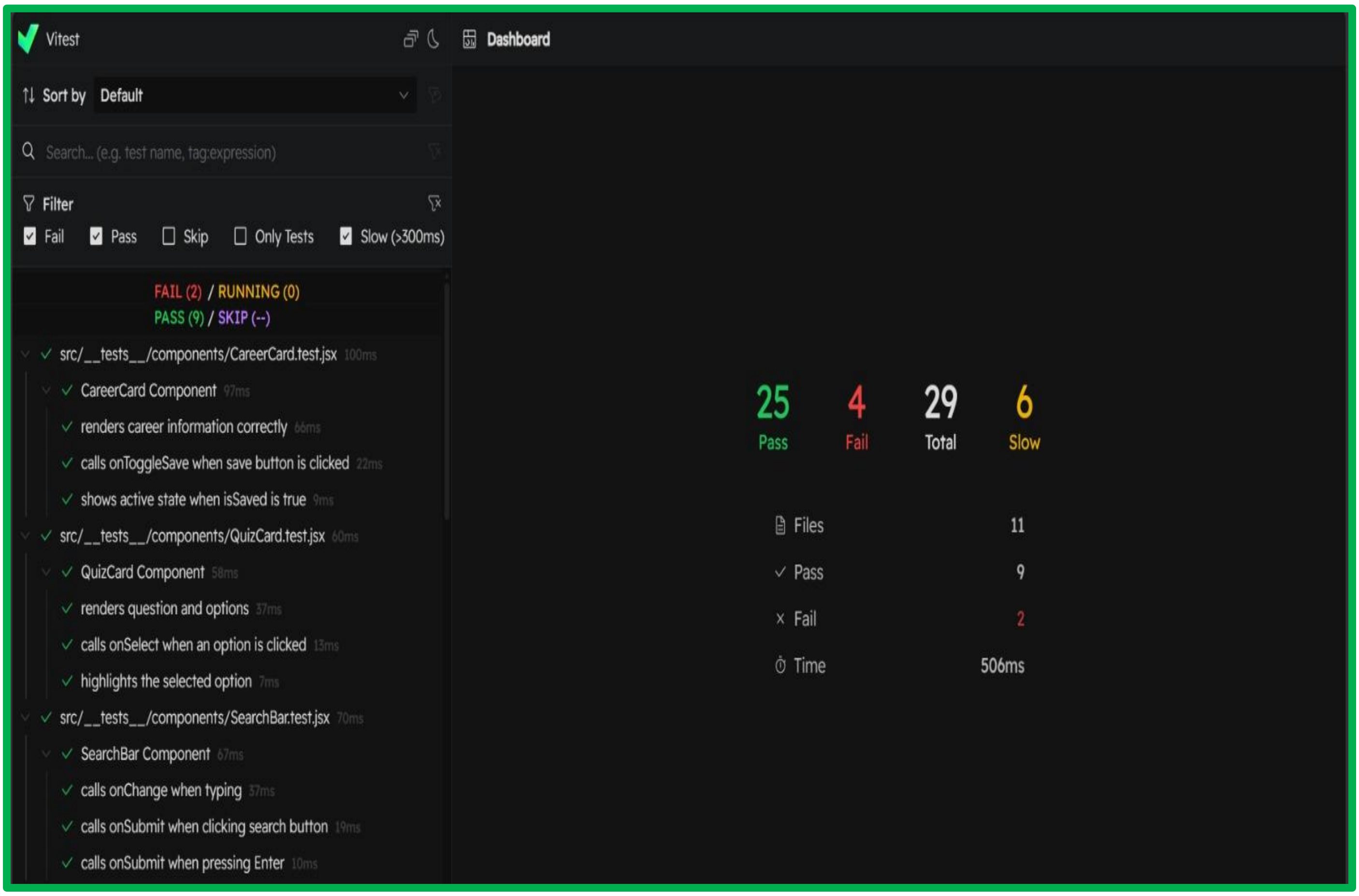
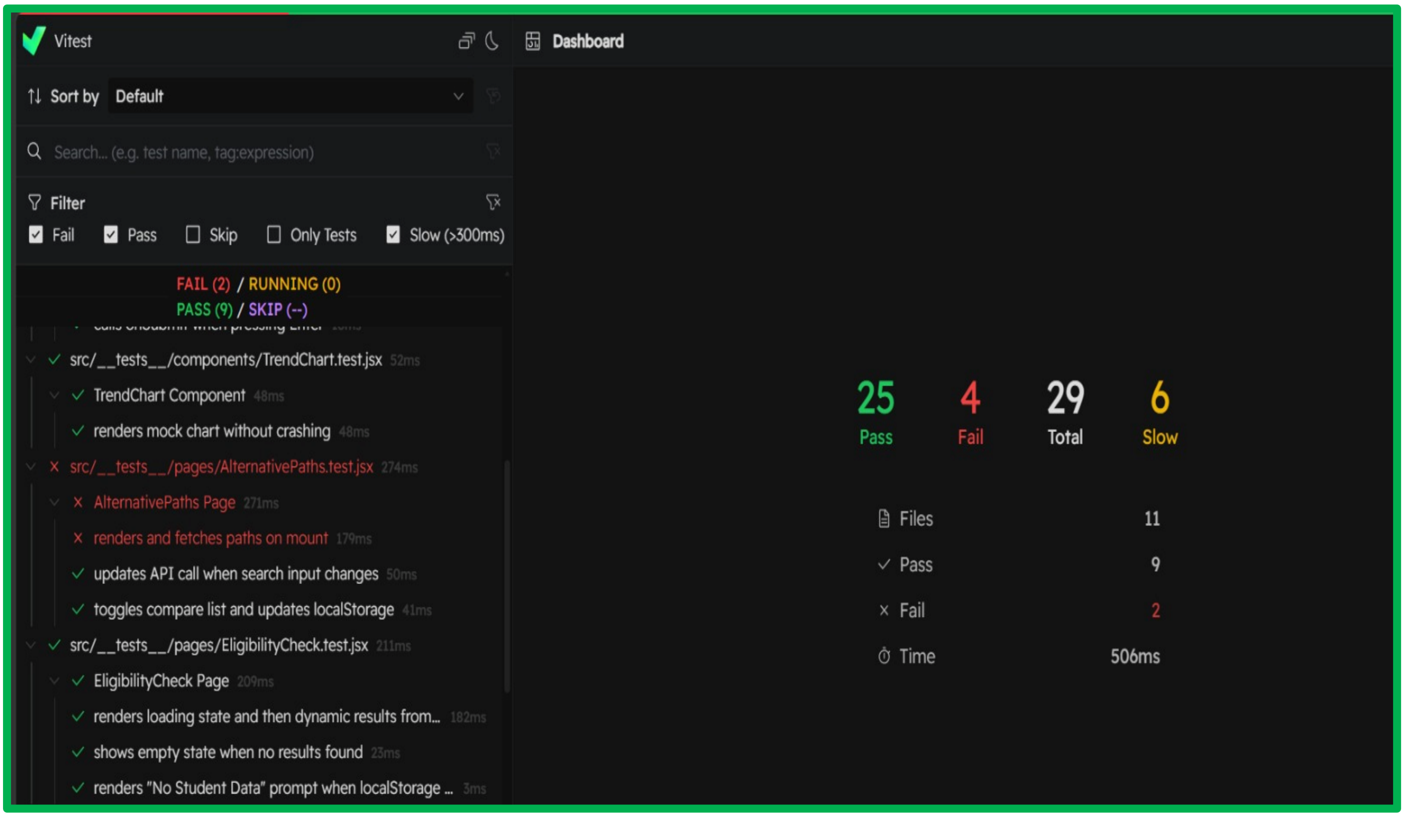


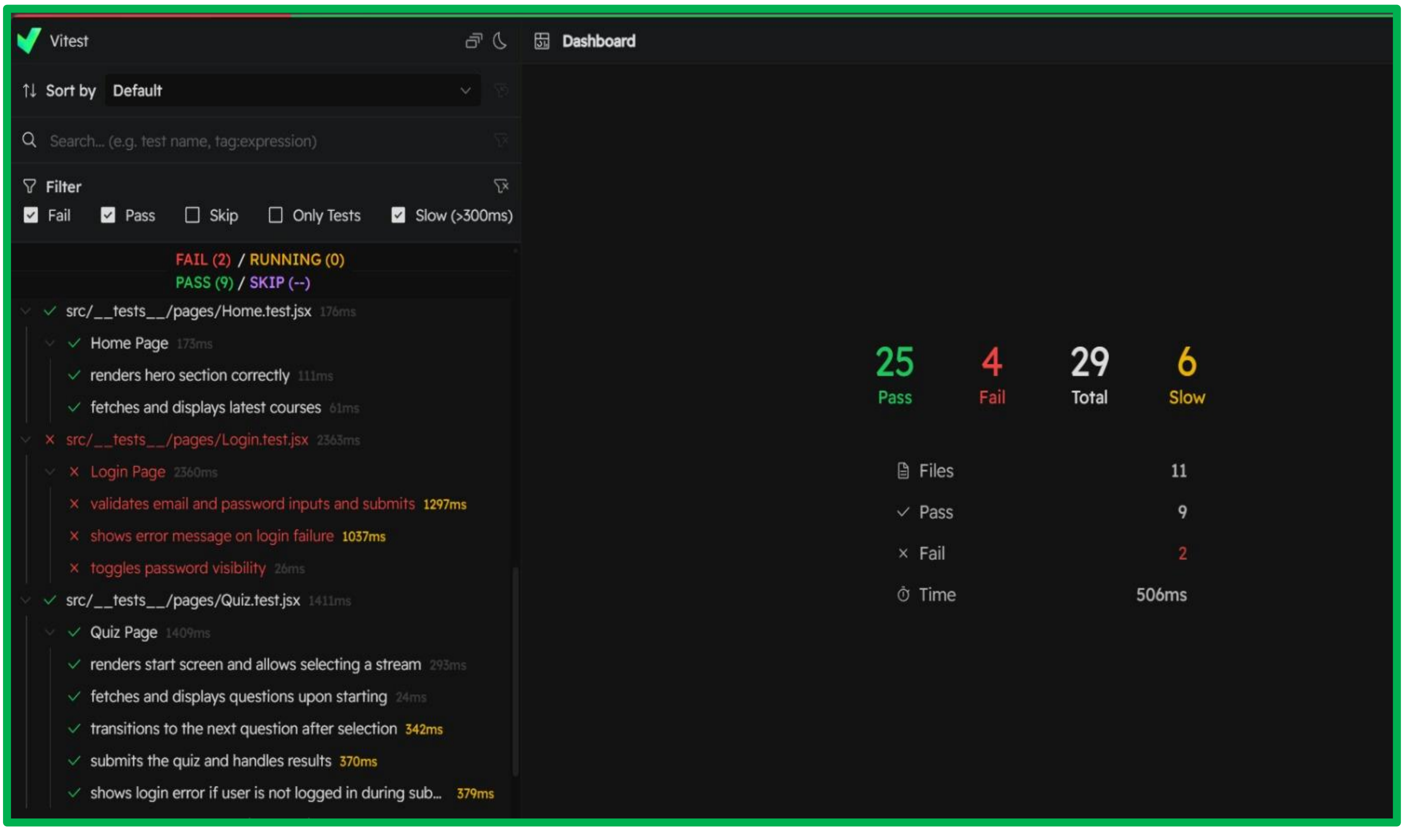
Figure B.6: List the eligibility criteria programmes

Figure B.6: Trend of past 6 years of eligibility pattern


Vitest
Dashboard
Sort by Default
Search... (e.g. test name, tag:expression)
Filter
Fail Pass Skip Only Tests Slow (>300ms)
FAIL (0) / RUNNING (0)
PASS (1) / SKIP (--)
src/__tests__/models/careerModel.test.js 20ms
Career Model Validation 18ms
should validate a valid career object 14ms
should require a title 2ms
should require at least one stream tag 1ms
should enforce enum for demandLevel 1ms
4 Pass
0 Fail
4 Total
Files 1
Pass 1
Time 509ms


Vitest
Dashboard
Sort by Default
Search... (e.g. test name, tag:expression)
Filter
Fail Pass Skip Only Tests Slow (>300ms)
FAIL (2) / RUNNING (0)
PASS (9) / SKIP (--)
src/__tests__/components/CareerCard.test.jsx 100ms
CareerCard Component 97ms
renders career information correctly 66ms
calls onToggleSave when save button is clicked 22ms
shows active state when isSaved is true 9ms
src/__tests__/components/QuizCard.test.jsx 60ms
QuizCard Component 58ms
renders question and options 37ms
calls onSelect when an option is clicked 13ms
highlights the selected option 7ms
src/__tests__/components/SearchBar.test.jsx 70ms
SearchBar Component 67ms
calls onChange when typing 37ms
calls onSubmit when clicking search button 19ms
calls onSubmit when pressing Enter 10ms
25 Pass
4 Fail
29 Total
6 Slow
Files 11
Pass 9
Fail 2
Time 506ms


Vitest
Dashboard
Sort by Default
Search... (e.g. test name, tag:expression)
Filter
Fail Pass Skip Only Tests Slow (>300ms)
FAIL (2) / RUNNING (0)
PASS (9) / SKIP (--)
src/__tests__/components/TrendChart.test.jsx 52ms
TrendChart Component 48ms
renders mock chart without crashing 48ms
src/__tests__/pages/AlternativePaths.test.jsx 274ms
AlternativePaths Page 271ms
renders and fetches paths on mount 179ms
updates API call when search input changes 50ms
toggles compare list and updates localStorage 41ms
src/__tests__/pages/EligibilityCheck.test.jsx 211ms
EligibilityCheck Page 209ms
renders loading state and then dynamic results from... 182ms
shows empty state when no results found 23ms
renders "No Student Data" prompt when localStorage ... 3ms
25 Pass
4 Fail
29 Total
6 Slow
Files 11
Pass 9
Fail 2
Time 506ms


Vitest
Dashboard
Sort by Default
Search... (e.g. test name, tag:expression)
Filter
Fail Pass Skip Only Tests Slow (>300ms)
FAIL (2) / RUNNING (0)
PASS (9) / SKIP (--)
src/__tests__/pages/Home.test.jsx 176ms
Home Page 173ms
renders hero section correctly 111ms
fetches and displays latest courses 61ms
src/__tests__/pages/Login.test.jsx 2363ms
Login Page 2360ms
validates email and password inputs and submits 1297ms
shows error message on login failure 1037ms
toggles password visibility 26ms
src/__tests__/pages/Quiz.test.jsx 1411ms
Quiz Page 1409ms
renders start screen and allows selecting a stream 293ms
fetches and displays questions upon starting 24ms
transitions to the next question after selection 342ms
submits the quiz and handles results 370ms
shows login error if user is not logged in during sub... 379ms
25 Pass
4 Fail
29 Total
6 Slow
Files 11
Pass 9
Fail 2
Time 506ms